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High-Frequency Measures of Informed Trading and Corporate Announcements

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ABSTRACT

We explore the dynamics of informed trading around corporate announcements of merger bids, dividend initiations, SEOs, and quarterly earnings by calculating *daily* posterior probabilities of informed buying and selling. We find evidence of informed trading before the announcements and a significant part of the news in announcements is impounded in stock prices before the announcements by pre-event informed trading. We also find evidence of informed trading after the announcements. Most strikingly, the probability of informed trading after merger bids predicts the probability of the bid being withdrawn or met with a competing bid. For other announcements, post-announcement informed-trading probabilities predict subsequent returns.

JEL Classification: G12

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The adjustment of security prices to the information of privately informed traders has been studied extensively in a literature starting from Grossman (1976), Grossman and Stiglitz (1980), Kyle (1985), and Easley and O'Hara (1987). While analytical research on this topic is well established, it remains a challenge to obtain empirical evidence on trading by informed investors because of the difficulty of determining when trades result from private information. Baruch, Panayides, and Venkataraman (2016) adopt an innovative approach to this problem that is designed for the detection of informed trading before unscheduled events such as mergers. They use order-placement data from the Euronext-Paris Exchange to proxy informed trading by the difference between order placements prior to the event and order placements during a control period. Other studies have examined trades around corporate events by particular classes of traders such as institutions or corporate insiders and from this evidence they have made inferences about informed trading. However, informed trading may result from the activities of much broader classes of traders, may occur before *scheduled* corporate events, and may occur *after* as well as before the events, and thus the challenge of identifying informed trading by unspecified traders in these circumstances remains. Indeed, Kim and Verrecchia (1994), Morris (1994), and Biais and Bossaerts (1998)¹ show that informed trading will take place after a public announcement whenever investors do not share common priors.

In this paper, we study informed trading both before *and* after three unscheduled corporate announcements (M&As, SEOs, and dividend initiations), as well as around pre-scheduled earnings announcements. We calculate the posterior probabilities of informed trading in a stock for each day, given the observed buy and sell transactions for that day. The probability measures are based on *PIN*, the probability of informed trading in the model of Easley, Kiefer, O'Hara, and Paperman (EKOP) (1996). We

¹Bond and Eraslan (2010) show that informational differences can lead to trade if an asset's owner makes decisions that affect that asset's value. Information revealed through trade is then socially valuable, which increases the gains from trade and allows trading to take place.

extend the *PIN* measure in several ways that allow us to characterize more precisely the behavior of informed traders. First, following Brennan, Huh, and Subrahmanyam (2016), we distinguish between informed selling and informed buying. This allows us to relate the direction of informed trading to subsequent favorable or unfavorable news announcements as well as to the direction of subsequent price movements. Secondly, by calculating the posterior probabilities of informed buying and selling each day, we are able to study the dynamics of informed trading around public announcements on a daily basis. Moreover, by conditioning on additional information (daily buys and sells), our measures can improve the precision of detecting informed-trading activities. While many previous studies of asset pricing and information asymmetry have used estimates of *PIN*, they have relied almost exclusively on unconditional estimates over multi-day (often annual) windows, which limit the reliability of the conclusions that can be drawn, and no study, so far as we are aware, relates probabilities of informed buying and selling to future corporate decisions such as new merger bids or bid withdrawals.

For unscheduled announcements, as well as scheduled earnings announcements, we find evidence of informed trading taking place both before and after the announcements. There are two kinds of evidence of informed trading prior to the announcements: first, a pre-announcement increase in the estimated probabilities of informed trading; and second, an association between the probabilities of informed trading in the pre-announcement period and the magnitude of the price reaction to the announcements. For M&A announcements, we find a big increase in the average probability of informed buying in target firms starting 25 trading days before the announcement and reaching 15% on the day before the announcement.² We also find increases in the average probabilities of informed buying (6%) and informed selling (3%) on the day before quarterly

²Cao, Chen, and Griffin (2005) examine informed trading in target firms around M&A announcements. They use signed volume (i.e., order imbalance) as a measure of informed trading in the options and stock markets, and find that the options market plays a more important role in price discovery.

earnings announcements. This contrasts with Benos and Jochec (2007), who are unable to detect increases in the probability in the pre-earnings-announcement period using an unconditional version of *PIN* that is estimated over 34 days on average.

Although there is little evidence of abnormal levels of informed trading on average prior to SEOs and dividend initiations, there is cross-sectional evidence that greater informed trading prior to all four types of announcements (M&As, SEOs, dividend initiations, and quarterly earnings) attenuates the stock-price responses to the announcements. This is consistent with the price adjusting to the new information through informed trading prior to the announcements.

Our evidence of informed trading after public announcements also takes two forms. First, the estimated probabilities of informed trading are abnormally high after the announcements. Secondly, the post-announcement probabilities of informed buying and selling predict both future (good and bad) events and subsequent returns. On the day after merger announcements, the average abnormal probability of informed buying of bidder firms is 16% and remains positive for up to 14 days; the average abnormal probability of informed buying of target firms is 10% on the day following the announcement, but declines rapidly and becomes negative four days after the announcement. However, the abnormal probability of informed selling of target firms is above 40% on the two days following the bid announcement and remains above 10% for eight days. For SEOs, the average abnormal probability of informed selling on the day following the announcement is around 6%, while for dividend initiations the average abnormal probability of informed buying is 8-11% on the two days following the announcement. For quarterly earnings announcements, the probabilities of both informed buying and informed selling are abnormally high on the days following the announcement, reaching 12% (6%) for informed buying (selling) on the day after the announcement.

Since it is possible that our estimates of the probabilities of informed trading after an-

nouncements are affected by factors that are not encompassed by the simple *PIN*-model,³ it is important to show that the estimates really do reflect trading on information. For this reason, we relate our estimates of the probabilities to subsequent events and returns. Merger announcements provide the richest opportunities for assessing whether the estimated probabilities of informed trading reflect information about future events, since each merger bid is followed by a merger outcome – success or withdrawal of the bid, or the emergence of a competing bid. If there is informed trading after a bid announcement, we expect it to be informative about the bid outcome.

To test this, we estimate probit regressions in which the dependent variable is a dummy set at unity if a bid is subsequently withdrawn, and the independent variables include the average probabilities of informed buying and selling of the target firm over the 10 days following the bid announcement. When we estimate the regression using the whole sample of target firms, the probabilities of informed trading are not significant. We conjecture that this is because, in stock-payment bids, informed trading is likely to be driven by merger-arbitrage trading that attempts to profit from discrepancies between the relative values of target and bidder shares and the proposed exchange ratio. When we restrict the sample to target firms of cash-only bids, we find that the probability of informed selling in the 10 days after the bid announcement is positively and significantly related to the probability of bid withdrawal; the probability of informed buying is negatively related to the probability of bid withdrawal but the relation is not significant.

A similar analysis for the emergence of a competing bid, which will tend to raise the final consideration received by target shareholders, shows that, for the full sample, the probability of informed buying is positively associated with the emergence of a competing bid, while the probability of informed selling is negatively related. When the sample is restricted to cash-only bids, the probability of informed buying remains significant while

³For example, order-flow shocks related to illiquidity. See Duarte and Young (2009).

the probability of informed selling becomes insignificant. These two sets of results suggest that our measures of informed trading after the bid announcement do indeed capture private information about the bid outcome.

To provide further evidence of the predictive power of post-bid informed trading probabilities, we regress the average daily return of the target shares, from day 11 following the announcement to the day on which the merger becomes effective or the bid is withdrawn, on the probabilities of informed trading in the target shares over the first 10 days following the bid announcement. For all-cash bids, the probability of informed buying is positively and significantly related to the subsequent return. We perform a similar analysis for a (much smaller) sample of all-stock bids in which the dependent variable is the average daily return difference between target and bidder shares up to the final date. Now the probability of informed buying in the target is significantly positively related to the subsequent return of the target firm shares relative to the return of the bidder firm shares, although the relationship becomes weak when a large set of control variables is included. The positive relation between informed buying of the target shares and the subsequent relative return is consistent with informed merger-arbitrage trading.

Overall, our evidence for post-bid informed trading is supportive of the hypothesis that informed trading takes place after public announcements. Our results provide a new perspective on the findings of Aktas et al. (2007), who report that *PIN* is higher after merger announcements than before and conclude from this that the *PIN* measure is defective. Our findings suggest that the post-announcement increase in their estimates of *PIN* is evidence, not of a defect in *PIN* itself, but of the greater importance of informed trading on *public* information in the post-announcement period, relative to informed trading on *private* information in the pre-announcement period.

We also find evidence of the predictive power of our post-announcement informed-trading probabilities for dividend initiations, SEOs, and quarterly earnings announce-

ments. The average probability of informed buying in firms during the 10 trading days after a dividend-initiation announcement is positively and significantly associated with the returns over the subsequent 50 trading days, and the average probability of informed selling in firms during the 10 trading days after an SEO announcement is negatively and significantly associated with the returns over the next 50 trading days.

For regular, scheduled announcements of quarterly earnings, we expect the price adjustment process to be more rapid, since analysts can prepare earnings-contingent valuations in advance of the announcement. Therefore, we relate the probabilities of informed trading computed over two days after the announcement to the subsequent returns up to 15 trading days after the announcement. We find that informed buying is significantly positively associated with the subsequent returns. For informed selling, the relation is negative but only marginally significant when the control variables are included in the regression. The finding of post-earnings-announcement informed trading is consistent with Krinsky and Lee (1996), who report an increase in the adverse-selection component of the bid-ask spread after earnings announcements, and with Green (2004), who finds a similar but more short-lived phenomenon in government bond markets after macro-economic news announcements.

While the evidence that the probabilities of informed trading derived from the EKOP (1996) model have predictive power for returns is consistent with the probabilities capturing informed trading, it is not necessarily consistent with the assumption of the *PIN* model that the market maker sets prices to reflect information in the past order flow or, more generally, with market efficiency. However, a closer look at the *PIN*-model structure reveals that the assumption about market-maker behavior and market efficiency is not crucial to the derivation of the empirical measures of informed trading. All that is required is that informed traders buy (or sell) when there is a discrepancy between the value of the security conditional on their information and the market price.

The basic *PIN* model structure of EKOP has been generalized to more complex versions elsewhere in the literature. For example, Easley et al. (2008) extend the basic EKOP (1996) structure to allow for time-varying arrival rates of informed and noise traders. We maintain the simpler setting of the EKOP (1996) model but focus on the daily *posterior* probabilities of informed trading and distinguish between informed buying and informed selling. Our sample contains more than 1,700 stocks each day, but the additional complexity of their model forces them to restrict their analysis to a sample of only 16 stocks. Similarly, Tay et al. (2009) propose a more complex model that accounts for time between transactions and also yields daily estimates of *PIN*, but they estimate it for only five companies for one year, presumably on account of computational costs. In estimating a model that incorporates the price impact of a trade as well as its direction, Back, Crotty, and Li (2016) find an increase in the conditional probability of an information event in the five days following an earnings announcement. Kumar and Popescu (2014) construct an intraday measure of the probability of informed trading that relies on bid/ask prices, market depth, and stock volatility, but does not distinguish between informed buying and informed selling.

In sum, our contribution is (i) to calculate metrics for informed buying and selling separately on a daily basis, (ii) to examine how they behave across a variety of scheduled and unscheduled corporate announcements, and (iii) to investigate whether these measures predict corporate outcomes and stock returns before and after the announcements. To the best of our knowledge, such a comprehensive investigation of informed-trading metrics around corporate events and their relation to future stock returns has not been conducted in the literature.

Collin-Dufresne and Fos (2015) (CF), in a significant paper, use 13(d) filings to show that insiders use limit orders to trade on “long-lived” (CF, p. 1563) information. Corporate announcements constitute perishable information, which may be short-lived and

necessitate the use of market orders, so that informed traders may use both limit and market orders depending on the nature of information. Thus, CF's point that informed agents use limit orders for long-lived information is complementary to our analyses.⁴ In addition to analyzing informed trading after as well as before corporate events, we also extend the sample of Baruch, Panayides, and Venkataraman (2016), which is limited to 101 corporate events. Our sample includes over 2,600 targets of merger bids, 720 dividend initiations, 1,710 SEOs, and earnings announcements for more than 1,410 firms per quarter for 31 years.

1. Daily Posterior Probabilities of Informed Trading

In the EKOP (1996) model of informed trading in an individual stock, one of three possible events occurs each day: there is no news ($\emptyset$), good news (g), or bad news (b). The unconditional probabilities of these events are denoted by $Pr(\emptyset) = (1 - \alpha)$, $Pr(g) = \alpha(1 - \delta)$, and $Pr(b) = \alpha\delta$, respectively, where α is the probability that an information event occurs on the day, and δ is the probability that the event is bad news. If an event occurs, it is observed only by a class of informed traders who trade to take advantage of it: if a good-news (bad-news) event occurs, the informed traders buy (sell) at the rate μ , and, whether or not a news event occurs, noise traders buy and sell at the rates ϵ_B and ϵ_S , respectively. The authors show how to estimate the model parameters from a time series of the numbers of daily buyer- and seller-initiated transactions. The unconditional probability of informed trading, PIN , is defined as the probability that a trade is initiated by an informed trader and is given by $\frac{\alpha\mu}{\alpha\mu + \epsilon_B + \epsilon_S}$.

In the spirit of the EKOP (1996) model, we develop the posterior probability that a given trading day was a no-news, good-news, or bad-news day, conditional on observing

⁴Indeed, in unreported analyses, we find that a measure of order imbalance based on the limit-order book plays a modest role in forecasting returns around corporate announcements.

the numbers of daily buyer-initiated trades (B) and seller-initiated trades (S) that day. Now, using Bayes' rule, the daily posterior probability that no information event has occurred on a given day, conditional on observing B and S , can be expressed as:

$$Pr(\emptyset|B, S) = \frac{Pr(B, S|\emptyset)Pr(\emptyset)}{Pr(B, S|\emptyset)Pr(\emptyset) + Pr(B, S|g)Pr(g) + Pr(B, S|b)Pr(b)} \quad (1)$$

Similar expressions can be derived for $Pr(g|B, S)$ and $Pr(b|B, S)$, the posterior probabilities that good-news and bad-news events have occurred.

Given the five parameters of the trading model, α , δ , μ , ϵ_B , and ϵ_S , it follows from the analysis of Easley et al. (1996) that these posterior probabilities are given by:

$$\pi(\emptyset|B, S) \equiv Pr(\emptyset|B, S) = \frac{(\alpha - 1)e^\mu \epsilon_B^B \epsilon_S^S}{\alpha(\delta - 1)\epsilon_S^S(\epsilon_B + \mu)^B - \epsilon_B^B[\alpha\delta(\epsilon_S + \mu)^S + (1 - \alpha)e^\mu \epsilon_S^S]} \quad (2)$$

$$\pi(g|B, S) \equiv Pr(g|B, S) = \frac{\alpha(\delta - 1)\epsilon_S^S(\epsilon_B + \mu)^B}{\alpha(\delta - 1)\epsilon_S^S(\epsilon_B + \mu)^B - \epsilon_B^B[\alpha\delta(\epsilon_S + \mu)^S + (1 - \alpha)e^\mu \epsilon_S^S]} \quad (3)$$

$$\pi(b|B, S) \equiv Pr(b|B, S) = \frac{\alpha\delta\epsilon_B^B(\epsilon_S + \mu)^S}{\epsilon_B^B[\alpha\delta(\epsilon_S + \mu)^S + (1 - \alpha)e^\mu \epsilon_S^S] - \alpha(\delta - 1)\epsilon_S^S(\epsilon_B + \mu)^B}. \quad (4)$$

For simplicity, we shall denote these posterior probabilities calculated each day by $\pi_\emptyset$, π_g , and π_b , respectively. Then, the posterior probability, conditional on observing B and S , that an information event has occurred on a given day is defined by $\pi_e = (1 - \pi_\emptyset)$. We refer to π_g (π_b) as the posterior probability of informed trading on good (bad) news, or the posterior probability of informed buying (selling). (More detailed procedure for computing the measures is available in the Internet Appendix.)

2. Data and Estimation of Model Parameters and Daily Posterior Probabilities

2.1. Data, Classification Algorithms, and Estimation

As a first step in estimating the daily posterior probabilities of informed trading, we process order flows using trades and quotes available from the Institute for the Study of Security Markets (ISSM) and the NYSE Trades and Automated Quotations (TAQ) databases. For the 1983-2006 period, the Lee and Ready (1991) algorithm is used to match trades with quotes and to classify each trade as buyer- or seller-initiated. For the period 1983 to 1998 we apply the five-second delay rule to match trades with quotes. Given the shorter reporting lag between trades and quotes in later years, we use the two-second-delay rule for the 1999-2006 period. Some issues related to applying the Lee-Ready method to the ‘monthly’ TAQ have been raised by researchers. Considering that we lag quotes when matching with trades, classification errors may not be serious for the 1983-2006 period, in which high-frequency-trading volume is relatively low.

We note, however, that the past decade has witnessed significant changes in regulation, market structure, trading technologies, and trading behavior of market participants. Stoll (2014), for example, documents that since the mid-2000s the number of trades per day has increased substantially while trade size has decreased, reflecting the increasing prevalence of high-frequency trading (HFT), especially since 2007. According to Arnuk and Saluzzi (2012), the introduction of the NBBO concept and Regulation NMS has made speed of execution paramount in the U.S. stock market, triggering a surge of HFT. Easley, Lopez de Prado, and O’Hara (2012) and Holden and Jacobsen (2014) suggest that applying the Lee-Ready (1991) method to the monthly TAQ database, which is time-stamped only to the second (as opposed to the millisecond), could induce substantial classification errors due to large HFT volume in recent years.

To alleviate the missclassification problem, Holden and Jacobsen (2014) propose a low-cost alternative, which is applicable to the monthly TAQ database, and show that their algorithm provides more accurate classifications than the Lee-Ready (1991) method. Therefore, we employ the Holden-Jacobsen algorithm for the last seven years in our sample (2007-2013).⁵ After matching trades and quotes based on either of the two algorithms, if a trade occurs above (below) the quote midpoint, it is considered buyer-initiated (seller-initiated). We limit our attention to NYSE/AMEX-listed stocks because transaction-level data for NASDAQ stocks are not available to us and the NASDAQ market has different trading protocols (Atkins and Dyl, 1997). Trades and quotes in the ISSM/TAQ databases that are out of sequence, recorded before the open or after the close, or involved in errors or corrections are excluded.

The five *PIN*-model parameters, α , δ , μ , ϵ_B , and ϵ_S , are estimated *monthly* via the Hwang et al. (2013) method using a three-month rolling window. The monthly estimation allows us to calculate the parameters that incorporate the time-varying features of information events in a firm and trading activities based on those events. Given the monthly estimates of the five parameters, the *daily* posterior probability estimates, $\pi_\emptyset$, π_g , and π_b , are then calculated from the numbers of buys (B) and sells (S) each day in the following month, using Equations (2)-(4). The procedure is repeated for the 369 months and 7,626 trading days from April 1983 to December 2013.⁶

The daily posterior probabilities are defined as follows:

$\pi_\emptyset \equiv \pi(\emptyset|B, S)$: The posterior probability conditional on the number of daily buyer-

⁵The main points of the Holden-Jacobsen (2014) algorithm are: (i) adjustments for withdrawn quotes, (ii) time-interpolation during each one-second period, (iii) matching trades with national best bid and offer (NBBO) quotes across different exchanges, and (iv) excluding crossed or locked NBBOs. So the time delay between quotes and trades is minimal (milliseconds at most). In addition, the number of trades in order flows processed by this algorithm increases substantially, compared to that processed by the Lee-Ready (1991) method.

⁶Since model parameters estimated over the previous three months are used to calculate the current month's posterior probabilities of informed trading, the daily posterior probability estimates start from the second quarter of 1983.

initiated trades (B) and seller-initiated trades (S) that an information event did not occur on a given day, as defined in Equation (2).

$\pi_g \equiv \pi(g|B, S)$: The posterior probability that a *good*-news event occurred on a given day conditional on B and S , as shown in Equation (3).

$\pi_b \equiv \pi(b|B, S)$: The posterior probability that a *bad*-news event occurred on a given day conditional on B and S , as shown in Equation (4).

For comparison purposes, statistics of the monthly estimates of the unconditional probabilities corresponding to the conditional probabilities $\pi_\emptyset$, π_g , and π_b are reported in Panel E of Table 1. The unconditional probabilities are defined as follows:

$(1 - \alpha)$: The unconditional probability that no information event occurs on a given day.

$\alpha(1 - \delta)$: The unconditional probability that a good-news information event occurs on a day (δ is the probability with which the information event contains bad news).

$\alpha\delta$: The unconditional probability that a bad-news information event occurs on a day.

It is possible that estimates of α , the probability that a private information event occurs, may be biased in recent years as a result of changes in order submitting practice and the disconnect between underlying orders and trades due to order splitting. For example, studies document that, owing to the development of computer technology, aggressive informed traders tend to split orders into multiple, smaller ones.⁷ This would leave the relative number of buys and sells the same, but the increase in the arrival rate of informed trades may possibly lead to an upward bias in estimates of α , which would in turn bias the estimates of the posterior probabilities of informed trading.

To investigate this issue, we compute the cross-sectional average of the five model parameters (α , δ , μ , ϵ_B , and ϵ_S) each month and plot the time series of the average

⁷See Chan and Lakonishok (1995), Keim and Madhavan (1995, 1996), Chordia, Roll, and Subrahmanyam (2011), Degryse, de Jong, and van Kervel (2013), Brogaard, Hendershott, and Riordan (2014), and Stoll (2014).

values in Figure 1. Figures 1(B) and 1(C) show that the average estimated arrival rates, μ , ϵ_B , and ϵ_S , which are related to trading frequencies, were increasing until early 2009, consistent with Vega (2006). In particular, Figure 1(B) shows that the average estimate of μ , the arrival rate of informed trades conditional on a good-news event, has increased since the late 1990's, and remains elevated after 2007, unlike the estimated arrival rates of uninformed trades, ϵ_B and ϵ_S , in Figure 1(C), which decrease to pre-crisis levels by 2013. Despite the high level of μ in recent years, Figure 1(A) shows that the average probability that an information event occurs on a given trading day, α , is stationary over the sample period without any discernible structural change, which is again consistent with Vega (2006) but not with Duarte, Hu, and Young (2015). The probability that a news event is bad, δ , rises from about 0.4 to 0.5 around the time of the recent financial crisis, while α , the probability that an event (of any kind) occurs, remains close to 0.3.

The other variables whose statistics are reported in Tables 1 and 2 are defined as follows. *N_QuoteUpdate*: the number of daily NBBO quote updates, which are changes in the prevailing intradaily best bid or offer price (BBO), or changes in the displayed depth for the BBO across all exchanges [following Conrad, Wahal, and Xiang (2015)]; *N_Trade*: the number of trades executed across all exchanges within each day; *R*: the daily stock return from CRSP; π_g^{DY} : the Duarte and Young (2009) (DY)-based daily posterior probability of trading on good news; π_b^{DY} : the DY-based daily posterior probability of trading on bad news; π_{ls}^{DY} : the DY-based daily posterior probability of a liquidity shock (unrelated to information asymmetry);⁸ *PSPR*: the daily proportional quoted spread, which is the average of intradaily proportional spreads (in %) of bid and ask quotes matched with trades [i.e., ((dollar spread)/(quote mid-point))*100], processed from ISSM/TAQ; *OIMB*: daily market-order imbalance (in %) [i.e., ((#BUY - #SELL)/(#BUY + #SELL))*100],

⁸We use the model of Duarte and Young (2009) (DY), which allows for liquidity shocks to estimate daily measures that correspond to our daily posterior probabilities. The details are provided in the Internet Appendix.

processed from ISSM/TAQ;⁹ *TURN*: daily share turnover, obtained from CRSP; *SIZE*: natural logarithm of daily market value (the daily stock price times the number of shares outstanding) (in \$million); and *BTM*: the book-to-market ratio (quarter-end book equity divided by market value of equity) of the most recent quarter, where the book equity is obtained from Compustat.

2.2. Descriptive Statistics

Table 1 reports time-series averages of daily and monthly statistics of the cross-sectional distributions of the posterior (or conditional) and prior (or unconditional) probabilities of informed trading, as well as other variables. To see the effect of HFT, we report the statistics for the two subperiods [non-HFT era (1983-2006) and HFT era (2007-2013)], in addition to those for the whole sample period (1983-2013). The average number of stocks included each day (or month) ranges from 1,684.0 to 2,132.8 for the whole sample period, from 1,725.3 to 1,914.5 for the non-HFT era, and from 1,266.8 to 3,140.2 for the HFT era, depending on the availability of each variable.

Panel A shows that the average number of daily quote updates ($N_QuoteUpdate$) for a stock is 19,843.3 for the whole period, but that it explodes from less than 1,400 in the non-HFT era to more than 81,000 in the HFT era. The mean (median) number of daily transactions, N_Trade , is 374.4 (149.7) prior to the HFT era, and as large as 7,950.2 (2,810.6) in the HFT era. These two variables, and especially N_Trade , are highly positively skewed.

The averages of the posterior daily (Panel B) and unconditional monthly (Panel E) probabilities are very close. The average posterior (unconditional) probability of no

⁹To compare the effects of market-order imbalance and limit-order imbalance, we also obtain limit-order data from LOBSTER.com, which provides the data from June 27, 2007. As in Chordia et al. (2017), daily limit-order imbalance, *LOIMB*, is defined and computed similarly to *OIMB* using the LOBSTER database for the 2007:07-2012:12 period.

information event is $\pi_\emptyset = 67.9\%$ ($(1 - \alpha) = 68.9\%$) in the whole period. The average posterior (unconditional) probability of informed buying is $\pi_g = 19.7\%$ ($\alpha(1 - \delta) = 18.6\%$), while the average posterior (unconditional) probability of informed selling is $\pi_b = 12.4\%$ ($\alpha\delta = 12.5\%$). The lower probabilities of informed selling are consistent with higher costs of trading on bad information, which may involve costly short sales (e.g., see Lamont and Thaler, 2003; Ofek, Richardson, and Whitelaw, 2004). The posterior probabilities exhibit more than twice the variability across stocks on a given day as the unconditional probabilities due to the additional information contained in the numbers of buy and sell orders each day. π_g , π_b , and their unconditional counterparts are positively skewed, implying that very high probabilities of informed trading are concentrated among a few stocks; and the range of estimated probabilities is from zero to one. Comparing the two subperiods in Panel B, we find that the average π_g falls from 21.2% in the non-HFT era to 14.7% in the HFT era, while the average π_b increases from 11.7% in the non-HFT era to 14.9% in the HFT era, which is greater than the corresponding value of π_g . We observe a similar pattern in the monthly unconditional measures in Panel E.

Panel C shows that the DY-based measures behave differently. During the whole sample period as well as the non-HFT era, the average probability of trading on good news (π_g^{DY}) is larger than that of trading on bad news (π_b^{DY}), but it is equal to π_b^{DY} during the HFT era. Especially, the average probability of informed trading on bad news (π_b^{DY}) decreases from 13.2% in the non-HFT era to 12.6% in the HFT era, which had the recent financial crisis.

Panel D shows that the average spread ($PSPR$) decreases substantially from 1.85% in the non-HFT era to 0.52% in the HFT era, averaging 1.54% for the whole sample period. Daily (market-)order imbalance ($OIMB$) is much smaller (-0.53%) in the HFT era than in the non-HFT era (13.32%), perhaps reflecting the effect of the recent financial crisis. Daily share turnover ($TURN$) increases sharply from 0.4% in the first subperiod to 1.0%

in the recent subperiod.

Panel F shows that the distributions of the three probabilities ($\pi_\emptyset$, π_g and π_b) are bimodal with most of the mass concentrated below 0.1 or above 0.9. On 60.7% of firm-days the probability that no information event occurs ($\pi_\emptyset$) is above 0.9, while on 26.1% of the days it is less than 0.1. On the remaining 13.2% of the days there is more uncertainty about whether an information event occurred. On 74.8% (84.3%) of the firm-days the probability that there is informed trading on good (bad) news is below 0.1, and on 15.1% (9.0%) of the firm-days the probability of informed buying (selling) is above 0.9. Thus there is a high (> 0.9) probability of informed trading on about 24% of the firm-days.

Table 2 reports statistics on the daily correlations between the probability estimates and other key variables. Panel A shows that on average the probability of no news, $\pi_\emptyset$, is negatively correlated with the probability of trading on good news, π_g (-0.72), and the probability of trading on bad news π_b (-0.54), while the correlation between π_g and π_b is only -0.18 . Since correlations can be misleading when the data are highly non-normal as seen in Table 1, Panel B presents a contingency table showing the joint distribution of the extreme values of π_g and π_b . On 61.1% of firm-days the probabilities of both good- and bad-news informed trading are below 0.1; 15.8% of them are good-news days and 9.0% are bad-news days, leaving only about 14% of the days in which both π_g and π_b fall into the intermediate range of 0.1-0.9. Thus, while a high probability of good (bad) news implies a low probability of bad (good) news, low probabilities of good and bad news tend to occur together (on no-news days).

The middle part of Panel A shows that the DY-based probability of informed buying (π_g^{DY}) is negatively correlated (-0.15) with the probability of informed selling (π_b^{DY}) and positively correlated (0.03) with the probability of a liquidity shock (π_{ls}^{DY}). π_b^{DY} is negatively correlated (-0.03) with π_{ls}^{DY} . The correlation between π_g and π_g^{DY} is 0.50 , and that between π_b and π_b^{DY} is 0.45 .

The lower part of Panel A shows that the daily return (R) is positively correlated (0.16) with π_g and negatively correlated (-0.08) with π_b . Order imbalance ($OIMB$) shows a similar but stronger relation with the two probabilities, correlations of 0.27 and -0.25 respectively. $TURN$ is positively correlated with both π_g and π_b , which is expected since both informed buying and informed selling increase turnover.

Panel C reports time-series averages of the monthly cross-sectional correlations between the unconditional probability estimates. It is the unconditional probability counterpart of the upper part in Panel A, and the patterns of the (average) cross-sectional correlations of the monthly unconditional probability estimates are similar to those of the daily conditional probability estimates reported in Panel A.

Figure 2 plots time series of the monthly averages of the two daily posterior probabilities. While there is considerable variation in the monthly averages, there is no apparent trend. In most months, the average π_b is smaller than the average π_g , but during the 2007-2013 period of financial crisis, π_b is often much higher, which is consistent with the mean value in the HFT era reported in Panel B of Table 1.

3. Informed Trading around Unscheduled Corporate Events: M&As, Dividend Initiations, and SEOs

Mergers are important (unscheduled) corporate events that have major consequences for value and control. Prior research shows that M&A announcements have significant effects on the share prices of the firms involved, especially the target firms, so that trading on private information related to M&As can result in large profits (e.g., see Cornell and Sirri, 1992; Chakravarty and McConnell, 1999; Mulherin and Boone, 2000; Andrade, Mitchell, and Stafford, 2001; Aktas, de Bodt, and Roll, 2004). It is not surprising therefore that there is evidence of information leakage and insider trading prior to the

M&A announcement date (M&AD). In a study by Meulbroek (1992), for example, most cases of insider trading detected and prosecuted by the SEC occurred before the M&AD. Keown and Pinkerton (1981) and Meulbroek (1992) present evidence that trading on private information prior to bid announcements is associated with abnormal returns.

Given the pronounced price movements and trading activities around the M&AD, merger bid announcements present an ideal opportunity to examine the behavior of the daily probabilities of informed trading. We compare informed trading around the M&AD with that around dividend initiation announcements (DID) and the filing dates for seasoned equity offerings (SEOD), events that are likely to have less dramatic economic consequences than merger bids.¹⁰

3.1. Behavior of Daily Probabilities around the M&AD, DID, and SEOD

To analyze the behavior of the estimated daily posterior probabilities of informed trading around the merger announcement date (M&AD: day d), M&A data are taken from SDC Platinum for tender offers (both completed and withdrawn) in the U.S. between 1983 and 2013 that have a transaction value of at least \$50 million. After requiring that both bidder and target firms be listed on the NYSE/AMEX so that the probabilities can be calculated, the final sample includes 7,172 bidder firms and 2,623 target firms.

Given the average premium of 36% paid for target firms' shares [Kengelbach and Roos (2011)] and their positive pre-MA&D abnormal returns, as well as anecdotal evidence of pre-M&AD insider trading, we expect to find evidence of informed buying of target shares during the pre-M&AD period. To capture abnormal levels of informed trading around the M&AD, we calculate abnormal probabilities of trading on good and bad news, π_g^{abn} and π_b^{abn} respectively, where the abnormal probability is the posterior probability (π_g or

¹⁰The average return across the day of, and the day following, the announcements, for our sample of merger bids is 10.7%, for dividend initiations is 0.522%, and for SEOs is -0.590%.

π_b) for a particular day minus the average of the corresponding daily probabilities over 20 pre-announcement trading days from $d-60$ to $d-41$. Figure 3 plots the averages of the daily abnormal posterior probabilities of informed trading around the M&AD.

Figures 3(A1) and 3(A2) show that most of the informed trading in bidder firms is post-M&AD informed buying (π_g^{abn}). The abnormal probability of informed buying (π_g^{abn}) is close to zero for bidder firms in the pre-M&AD period, about 12% on the M&AD, and over 15% on day $d+1$, before declining rapidly over the following few days. For target firms in Figures 3(B1) and 3(B2), most of the pre-announcement informed trading is informed buying, as one might expect. In Figure 3(B1) we find that the abnormal probability of informed buying (π_g^{abn}) is positive from day $d-25$. It starts to rise around day $d-10$, and is over 5% on each of the following days, reaching 16% on day $d-1$. The abnormal probability (π_g^{abn}) remains positive for four days beyond the M&AD and then turns negative. Figure 3(B2) shows that the abnormal probability of informed selling (π_b^{abn}) of target firms jumps from 2% on day $d-1$ to over 37% on the M&AD and 47% on day $d+1$, declining towards zero over the next 14 trading days. The evidence shown above is partly consistent with Aktas et al. (2007), who report higher values of *PIN* in the post-M&AD period.¹¹

In Figures 4 and 5, we present similar plots around the announcement dates of two other unscheduled events: the dividend initiation date (DID) and the seasoned equity offerings date (SEOD). DID is the date on which a cash dividend is declared for the first time in a firm's history. The dividend initiation sample of 726 firms includes firms that pay ordinary quarterly dividends only (distribution code = 1234 in Compustat). SEOD is the filing date of SEOs in the Global New Issues database of SDC Platinum. The sample of 1,648 includes all offerings of at least \$25 million that issue at least some

¹¹The authors use very long windows to obtain three point estimates of *PIN* over each of the three intervals around the M&A announcement date, and they do not consider the 7 trading days ($-5, +2$) around the announcement "because the window is too short to infer values for the different *PIN* parameters" (see p. 175 of their paper).

primary shares.

Figure 4(A) shows that the average abnormal probability of informed buying, π_g^{abn} , is almost always positive in the 25 days preceding a dividend initiation announcement, but it is always less than 2% and is comparable to the corresponding probability of informed selling shown in Figure 4(B). However, it jumps on the announcement day and reaches 12% on day $d+1$, declining towards zero over the next 10 trading days. The abnormal probability of informed selling, π_b^{abn} , is positive and somewhat higher after than before the DID, but on no day reaches as high as 4%.

Evidence of abnormal probabilities of informed trading before SEOs, a bad-news event, is also slight. The abnormal probabilities of informed selling in Figure 5(B) are positive on almost every day but very small, and the corresponding probabilities of informed buying in Figure 5(A) are negative on several days but very small. On the SEOD both π_g^{abn} and π_b^{abn} jump. While π_g^{abn} falls back on day $d+1$, π_b^{abn} increases further to 7% and remains positive but small over the following 15 trading days.

Overall, the abnormal probabilities of informed trading around the DID and the SEOD are much lower than around the M&AD, and it is not clear that we have uncovered any evidence of unusual pre-announcement informed trading using the average abnormal probabilities of informed trading for these events. This is consistent with the bigger price reaction to M&A announcements which creates greater incentives for informed trading before M&A announcements than before SEO and dividend initiation announcements (See footnote 10). The smaller *average* abnormal probabilities of informed trading for SEOs and dividend initiations, however, do not preclude the possibility that *cross-sectional* variation in the probability estimates is related to cross-sectional variation in price adjustments and therefore in subsequent returns. In the following section, we examine whether pre-announcement informed trading causes adjustment of the stock price to the new information before it is publicly announced.

3.2. Pre-Announcement Informed Trading around the M&AD, DID, and SEOD, and Announcement Returns

Models of price formation in security markets (e.g., Kyle, 1985) suggest that informed trading causes the information on which it is based to be impounded in the stock price, attenuating the price response to a subsequent public announcement of the information. In the case of positive news we expect informed buying before the announcement to reduce the announcement return, and in the case of negative news we expect informed selling before the announcement to increase the announcement return. The informed trading hypothesis is confirmed by Meulbroek (1992) who shows that, for a sample of firms in which insider trading was charged by the SEC, almost half the pre-announcement run-up in the share price occurred on days on which insiders traded.

To investigate the informed trading hypothesis for merger bids, we regress the M&A announcement return on the average probabilities of informed buying and informed selling computed over 20 days prior to the M&AD: $\pi_g(-20, -1)$ and $\pi_b(-20, -1)$. The M&A announcement return, $CAR(0, +1)$ or $CAR(-1, +1)$, is the cumulative abnormal return over the two or three trading days (days d or $d-1$ to $d+1$) around the M&AD, where the abnormal return is the difference between the daily stock return and the S&P500 Index return. The results are reported in Panels A1 and A2 of Table 3. To control for the effects of other variables in all regression analyses, we include the average values of the variables over the same 20 days prior to the announcement. We include the past return [$R(-20, -1)$], return volatility [$RVOLA(-20, -1)$], the proportional spread [$PSPR(-20, -1)$], order imbalance [$OIMB(-20, -1)$], share turnover [$TURN(-20, -1)$], firm size [$SIZE(-20, -1)$], and the book-to-market ratio [BTM].¹²

Regression (i) in Panel A1 of Table 3 shows that the coefficient of $\pi_g(-20, -1)$ is

¹²The book-to-market ratio is not available on a daily basis, so BTM in all regressions is the ratio from the most recent quarter.

negative and significant so that, consistent with the informed trading hypothesis, the greater is the estimated probability of informed buying, the lower is the announcement return. The coefficient implies that a target firm experiencing pre-M&AD informed trading $[\pi_g(-20,-1)]$ higher by one standard deviation (0.219) is expected to observe an announcement return $[CAR(0,+1)]$ lower by 3.64%. However, it is possible that π_g has nothing to do with informed buying but is associated with positive price changes because of the way in which transactions are classified as buys.¹³ The estimate of π_g would then be associated mechanically with the pre-announcement price run-up, which could lead to a spurious relation between $\pi_g(-20,-1)$ and the announcement return, $CAR(0,+1)$. To allow for this possibility, regression (ii) includes the average stock return over the 20 days prior to the bid announcement, $R(-20,-1)$, as well as the other control variables. While the coefficient of the simple run-up variable, $R(-20,-1)$, is negative and significant, the coefficient of $\pi_g(-20,-1)$ remains highly significant. This is evidence that the price movement that is due to pre-M&AD informed trading has an incremental effect in reducing the price reaction to the announcement ('attenuation'), in addition to the effect of the general pre-announcement price change.

On the other hand, there is no evidence that pre-announcement informed selling, captured by $\pi_b(-20,-1)$ in specifications (iii)-(vi), has any effect on the announcement return. This is as we should expect, since any informed selling prior to the bid will be based on information other than the likelihood of a forthcoming bid. These findings not only validate π_g as a measure of informed buying, but also imply that information about tender offers leaks out prior to the M&AD and is partially impounded in the stock price through informed buying. We note also that return volatility, $RVOLA(-20,-1)$, is positively related to the announcement return, while $OIMB(-20,-1)$, $TURN(-20,-1)$, and $SIZE(-20,-1)$ are all negatively related to it. When $CAR(-1,+1)$ is used

¹³Table 2 shows that the average cross-sectional correlation between π_g and the stock return is 0.16.

as the dependent variable in Panel A2, the patterns are similar.

In Panels B and C of Table 3, we report the results of similar analyses around the dividend initiation date (DID) and the SEO date (SEOD). Panel B shows that the probability of pre-DID informed buying, $\pi_g(-20, -1)$, is negatively related to the announcement return, which is also consistent with the informed trading hypothesis since dividend initiations are good news. The coefficient of $\pi_g(-20, -1)$ is significant at the 1% level, and although the significance level falls to 10% when the seven control variables are included, none of the controls including the run-up variable, $R(-20, -1)$, is individually significant. Regression (i) shows that a one-standard-deviation increase (0.195) in $\pi_g(-20, -1)$ reduces $CAR(0, +1)$ by 0.53%. The coefficient on $\pi_b(-20, -1)$ is positive and significant at the 10% level except when the controls are included.

By contrast, SEOs are bad news (e.g., Loughran and Ritter, 1995), and the informed trading hypothesis predicts that informed selling before the announcement will increase the announcement return. That is what we see in Panel C. The coefficient of $\pi_b(-20, -1)$ is positively significant at 5% in the regressions without controls and at 1% when the controls are included. As one would expect, however, the attenuation effect around the SEOD is smaller than that around the M&AD: in regression (iii), for example, a one-standard-deviation increase (0.149) in $\pi_b(-20, -1)$ causes $CAR(0, +1)$ higher by 0.28%. There is no evidence that informed buying before the SEOD has any effect on the announcement return: the coefficient of $\pi_g(-20, -1)$ is insignificant in all specifications.

In sum, pre-announcement informed buying reduces the announcement return for positive news announcements (merger bids and dividend initiations), while informed selling increases the announcement return for the negative news represented by SEO announcements. Further consistent with the informed trading hypothesis, there is no evidence that informed selling before positive news announcements or informed buying

before negative news announcements affects the announcement return.¹⁴

3.3. Post-M&AD Informed Trading and Bid Outcomes

A merger bid is only the first step in what can be a complex acquisition process that may not end in a completed merger. The initial bid may elicit competing bids from rivals who fear the consequences of having a large competitor, or the bid may be withdrawn by the bidder because of regulatory concerns, anti-trust actions, changes in tax regulations and market conditions that make the merger no longer attractive, or defensive actions by the target firm such as a poison pill provision. In the first four months of 2016, for example, \$400 billion worth of corporate merger offers in the U.S. were withdrawn: these included proposed mergers of Staples and Office Depot, of Halliburton and Baker Hughes, and of Pfizer and Allergan, which were withdrawn because of regulatory concerns or changes in tax rules.¹⁵ The withdrawal of a merger bid can impose significant costs on investors in the target firm, including ‘arbitrage’ investors who are trying to profit from the difference between the market price of the target and the bid price: for instance, Allergan’s stock price dropped by 20% when hopes for the merger completion were dashed by changes in tax rules. On the other hand, the emergence of competing bidders can lead to a bidding war in which the final bid price is considerably above the initial price.

Therefore, once an initial bid has been made, investors have a strong incentive to assess the likelihood of a competing bid emerging or of the initial bid being withdrawn.

If some investors have a comparative advantage in making such assessments and trade

¹⁴For a robustness check, we use $CAR(-1, +1)$ and $CAR(0, +2)$ as the dependent variable in the panels of Table 3. The results (reported mostly in the Internet Appendix) are qualitatively similar. We have also repeated the analysis in this section using the DY-based measures, $\pi_g^{DY}(-20, -1)$ and $\pi_b^{DY}(-20, -1)$ [as well as $\pi_{ls}^{DY}(-20, -1)$ as a control variable]. The results with $\pi_g^{DY}(-20, -1)$ and $\pi_b^{DY}(-20, -1)$ are not consistent with those reported in Table 3. To compare the effect of limit-order imbalance, we also control for $LOIMB(-20, -1)$ in Panels A and C in Table 3. Although the number of observations decreases significantly because of limited availability of the LOBSTER database, the results are qualitatively similar. These results which are not reported here are available upon request.

¹⁵See the *New York Times*, May 12, 2016.

on their information, this will give rise to post-announcement informed trading, which is suggested by the abnormally high probabilities of informed trading that we observe after bid announcements. We therefore test the post-announcement informed trading hypothesis that the probabilities of informed trading in the post-M&AD period are informative about the later withdrawal of the initial bid or the appearance of other competing bid(s).

Out of the 2,623 target firms that received initial bids between 1983 and 2013, 633 bids were withdrawn and the remaining bids (1,990) were successful. To assess whether the post-announcement probabilities reflect information about possible withdrawal of the bid, we estimate probit regressions in which the dependent variable is equal to unity if the bid is withdrawn and zero otherwise. The explanatory variables are the average probabilities of informed buying and selling (and the averages of the control variables) over days +1 to +10 in the post-M&AD period. The results are reported in Table 4. Panel A includes all target firms, while Panel B includes only target firms that receive all-cash bids; out of the 1,008 sample firms in Panel B, 289 bids were withdrawn.

Panel A shows a negative but only marginally significant relation between the probability of bid withdrawal and the post-M&AD probability of informed buying, $\pi_g(+1, +10)$, and no significant relation between bid withdrawal and the probability of informed selling, $\pi_b(+1, +10)$. These results offer no support for the informed trading hypothesis. A possible reason for these findings is that the sample includes bids that offer stock, cash, or other consideration. For stock-payment bids, informed trading will depend not only on the probability of bid success but also on the ratio of the prices of the target and bidder shares relative to the proposed exchange ratio, and this relative-arbitrage activity may obscure the link between informed trading and bid withdrawal.

Therefore, in Panel B we restrict the sample to the target firms that receive all-cash bids. We still find some evidence in regression (iii) that a lower level of informed buying is associated with bid withdrawal. More importantly, now there is a highly significant

relation between informed selling as measured by $\pi_b(+1, +10)$ and bid withdrawal in regression (iii), which is robust to the inclusion of the control variables in regression (iv). This is consistent with the post-announcement informed trading hypothesis and suggests that the high probability of informed selling that we observe *after* the announcement in Figure 3(B2) reflects information about the subsequent withdrawal of the bid. This is also consistent with our hypothesis that informed trading can occur in the post-bid period, because some traders are able to interpret better and more rapidly the implications of the public information contained in the bid announcement.

While the withdrawal of a bid has negative consequences for the shareholders of the target firm, the emergence of a competing bidder is generally positive for them, since the new bid will typically be at a higher price than the initial bid and may force the original bidder to raise its price. Therefore, we also consider whether post-M&AD informed trading reflects information about the possible future emergence of competing bids. We use data from SDC Platinum to determine whether each initial bid is followed by one or more competing bids and estimate a probit model, in which the dependent variable is equal to one if a competing bidder emerges after the initial bid and zero otherwise; the independent variables are the probabilities of informed trading after the initial bid as in the previous table. [Out of the total 2,558 initial bids, the first bidder in 334 M&A cases faces at least one competing bidder after the initial bid.] The results are reported in Table 5. Panel A includes all target firms. To avoid any confounding effect of relative-value merger arbitrage trading referred to above, in Panel B the sample is limited to the targets that receive all-cash bids; this reduces the number of observations to 964.

For both the full sample and the subsample of targets that receive all-cash bids, the coefficient of $\pi_g(+1, +10)$ is positive and significant at the 5% level or better in all the regressions. This evidence that post-announcement informed buying reflects information about the probability of a competing bid emerging is consistent with the post-

announcement informed trading hypothesis. The coefficient of $\pi_b(+1, +10)$ is negative and significant at the 5% level for the full sample (Panel A), but is generally insignificant for the smaller sample (Panel B). A lower intensity of informed selling when the probability of a competing bid is high is consistent with, but not implied by, the informed trading hypothesis. Overall, our results for the emergence of new bids are also consistent with the post-announcement informed trading hypothesis.

3.4. Post-Announcement Informed Trading and Returns After the M&AD, DID, and SEOD

To provide further evidence on the post-announcement informed trading hypothesis, we test whether the post-announcement probabilities predict subsequent stock returns. First, for merger bid announcements we regress the target share return following the M&AD on the informed trading probabilities for the target firm averaged over the 10 post-M&AD trading days, $\pi_g(+1, +10)$ and $\pi_b(+1, +10)$, as well as the control variables. The dependent variable in these regressions, $R(+11, \text{FND})$, is the average of daily target firm stock returns between day $d+11$ and the final date, FND, which is the effective date for completed bids and the withdrawal date for withdrawn bids.

The results are reported in Panels A1 and A2 of Table 6. Panel A1 shows that, for the sample of all target firms, the informed trading probabilities have no predictive power for future stock returns. However, as mentioned above, bids that include a stock element may induce informed trading that is motivated by differences between the relative prices of the bidder and target shares rather than by the absolute return prospects of the target. Panel A2, where the sample is limited to all-cash bids, shows that for these bids the average probability of informed buying, $\pi_g(+1, +10)$, is significantly positively related to the subsequent return, $R(+11, \text{FND})$. From the perspective of economic significance, a one-standard-deviation move in $\pi_g(+1, +10)$ increases $R(+11, \text{FND})$ by 0.1%. That is,

more intensive informed buying in the period after the announcement predicts higher future stock returns in target firms.

We have suggested that relative-value merger-arbitrage trading in bidder and target shares following stock bids may obscure the relation between informed trading and future returns for the target firms. If such arbitrage trading is important, we might expect informed trading in the target shares following stock-bid announcements to predict the *relative* returns to bidder and target shares. To examine this possibility in Panel B of Table 6, we limit the sample to target firms that receive all-stock bids only. We compute the difference of daily returns between each target firm and its bidder firm ($R_{Target} - R_{Bidder}$) in the post-M&AD period, and obtain the average of return differences from day $d+11$ to the final date, $DR(+11, FND)$. To obtain the return differences, each NYSE/AMEX-listed target firm is matched with its bidder firm listed on the NYSE/AMEX, which reduces the sample size to only 228 bids. We then regress $DR(+11, FND)$ on the probabilities averaged over the 10 days after the M&AD. Panel B shows that the coefficients of the informed buying variable, $\pi_g(+1, +10)$, are positive in the two regressions and statistically significant in regression (iii), which does not include the control variables. While these results lend only mild support to the hypothesis of relative-value arbitrage, we note that the sample size in these regressions is relatively small.

We estimate similar regressions for dividend initiations as well as for SEOs and report the results in Table 7. The dependent variable is the average of daily stock returns from days $d+11$ to $d+60$ (or $d+30$) following the announcement [$R(+11, +60)$ or $R(+11, +30)$]. The primary independent variables are the post-announcement informed trading probabilities, $\pi_g(+1, +10)$ and $\pi_b(+1, +10)$. Panel A shows that the post-DID probability of informed buying, $\pi_g(+1, +10)$, predicts a higher subsequent stock return, $R(+11, +60)$. For the bad-news event in Panel B, we also find that the post-SEOD probability of informed selling, $\pi_b(+1, +10)$, predicts a lower level of stock returns. The

results are statistically significant and robust to including the control variables or using $R(+11, +30)$ (in Panels C and D). These findings confirm that the post-announcement probabilities of informed trading are informative about subsequent returns.¹⁶

4. Informed Trading around a Pre-Scheduled Corporate Event: Earnings Announcements

Earnings announcements provide another major opportunity for informed trading, because they are frequent and can have major valuation implications. Since earnings announcement dates (EAD) are typically known in advance, a trader with private information faces a firm deadline for exploiting that information, which could lead to an increase in informed trading as the announcement date approaches (Foster and Viswanathan, 1996). Moreover, unlike the announcements we have considered thus far, earnings announcements can be categorized into either good news or bad news in a straightforward fashion via earnings surprises, which allows us to analyze the two cases separately. We consider first the behavior of the informed trading probabilities around the EAD, and then use them to provide further evidence on the informed trading hypothesis. Previous studies (e.g., Benos and Jochev, 2007; Brown, Hillegeist, and Lo, 2009) have analyzed the behavior of PIN around earnings announcements using coarser measures than ours.

4.1. Behavior of the Daily Probabilities around the EAD

To examine the behavior of the informed trading probabilities around the EAD, data are taken from the CRSP/Compustat Merged file and the IBES database. If the quarterly earnings were announced after the close (4:00 p.m. EST) on the IBES-reported announcement date, which is indicated by the time-stamps (available from 1999), the

¹⁶We have also estimated regressions similar to those reported in Tables 4-7 using the DY-based probabilities. The results which are available on request are not consistent with those reported here. To compare the effect of limit-order imbalance, we also control for $LOIMB(+1, +10)$ in Panel B of Table 7 and the results are consistent with those in Table 7.

announcements are assumed to be made on the following trading day. To capture the behavior of informed trading around the EAD, we again compute the averages of the abnormal probabilities of informed trading on good and bad news, π_g^{abn} and π_b^{abn} , respectively, where the abnormal probability is measured by the difference between the probability (π_g or π_b) for a particular day and the average of the corresponding daily probabilities over 30 pre-EAD days from $d-40$ to $d-11$. Recent studies document that 43%-45% of quarterly earnings announcements are made after the close (4:00 p.m. to midnight).¹⁷ If earnings are announced after the close in the period 1983-1998 for which time-stamp data are not available, the news will not be reflected in the stock price of day d , so that informed trading on this day would still be based on *private* information.

Figure 6 plots the average of π_g^{abn} and π_b^{abn} around the EAD. There is no sign of abnormal levels of the probabilities prior to day $d-2$. In Figures 6(A) and 6(B), the abnormal probabilities start to increase on day $d-2$ and peak on the announcement date. On day $d-1$, the probability of informed buying in Figure 6(A) rises by about six percentage points, and the probability of informed selling in Figure 6(B) rises by about three percentage points. On day d itself, the probability of informed buying is about 17 percentage points higher than normal, and that of informed selling is about nine percentage points higher.¹⁸ The higher probabilities for good news than for bad news informed trading are consistent with lower costs of exploiting good news.

After the announcement, the abnormal probabilities remain positive but decline towards zero over the following 10 trading days. The raised post-announcement probabili-

¹⁷Michaely, Rubin, and Vadrashko (2014, 2016) show that only 36%-38% of quarterly earnings announcements are made before the open (midnight to 9:30 a.m. EST) on the EAD. The majority of announcements are made *during* or *after* the regular trading hours on the EAD: about 18% during the trading hours (9:30 a.m. to 4:00 p.m.) and 43%-45% after the close (4:00 p.m. to midnight).

¹⁸We cannot tell how much of the pre-announcement informed trading takes place before the announcement on the EAD (day d) for the reason mentioned above. When we plot the equivalents of Figure 6 separately for the non-HFT era (1983-2006) and the HFT era (2007-2013), the values of π_g^{abn} and π_b^{abn} in the pre-EAD period are generally smaller in the HFT era than in the non-HFT era, which is consistent with Weller (2017).

ties are consistent with what we have already observed for the targets of merger bids and for dividend initiations. The probabilities are below those we observed for M&A targets but above what we observed for dividend initiations and SEOs.

4.2. Earnings Surprises and the Probability of Pre-Announcement Informed Trading

Ceteris paribus, we should expect that information whose public release would trigger a big price reaction to offer a bigger incentive to trading by privately informed investors. It is natural therefore to relate the probabilities of pre-announcement informed buying and selling to the magnitude of the stock-price reaction to the subsequent public announcement. However, we have seen that for unscheduled announcements the probability of pre-announcement informed trading attenuates the price reaction to the public announcement, potentially destroying any relation between the announcement return and prior informed trading. Therefore, we relate the pre-EAD informed trading probabilities, not to the endogenously determined announcement return, but to the standardized unexpected earnings (SUE), or earnings surprise, which is unaffected by pre-EAD informed trading. Following Livnat and Mendenhall (2006), SUE is defined by $SUE_{it} = \frac{EPS_{it} - \Psi}{P_{it}}$, where EPS_{it} is the earnings per share for firm i in quarter t , P_{it} is the stock price at the end of quarter t , and Ψ is the EPS in quarter $t-4$ (i.e., EPS_{it-4}) [adjusted for stock splits and stock dividends, as in Abarbanell and Lehavy (2007)]. Thus, the quarterly earnings surprise is computed by assuming that EPS follows a seasonal random walk. The advantage of this definition is that quarterly earning surprises can be estimated for almost all firms, unlike other SUE definitions that require analysts' forecasts.

To determine whether the probability of informed trading before the EAD is associated with the earnings surprise, the averages of the conditional probabilities over the five trading days prior to the EAD, $\pi_g(-5, -1)$ and $\pi_b(-5, -1)$, are first calculated each quar-

ter for each stock. Then each quarter the stocks are assigned to ten equal-sized portfolios by sorting on *SUE*. In each portfolio in each quarter, the cross-sectional mean of the average probabilities obtained above is computed, and finally the time-series average of the quarterly cross-sectional means is reported in Table 8. The difference in the average probability of informed buying, $\pi_g(-5, -1)$, between the most positive (SUE10) and most negative (SUE1) surprise portfolios is 4.4% with a *t*-statistic of 10.27; this compares with the average π_g of 19.7% shown in Panel B of Table 1. Similarly, the average probability of informed selling, $\pi_b(-5, -1)$, is 6.7% larger for the most negative *SUE* portfolio than for the most positive *SUE* portfolio, with *t*-statistic on the difference of -18.71; this compares with the average π_b of 12.4% reported in Table 1.

The above finding is evidence of informed trading on private information prior to the EAD: buying before positive earnings surprises and selling before negative earnings surprises.¹⁹ This is consistent with the findings of Brennan, Huh, and Subrahmanyam (2016) who show that, when *PIN* is estimated quarterly, its good-news component (*PIN_G*) is increasing in the earnings surprise and its bad-news component (*PIN_B*) is monotonically decreasing. Our focus, of course, is on the increased precision of the daily conditional measures of informed trading, which allows us to shed light on the relation between informed-trading measures and returns surrounding the announcement, an issue we will examine in Subsections 4.3. and 4.4..

Further evidence of informed trading on the days around the EAD is provided in Table 9, which reports the proportion of firms in the highest (SUE10) and lowest (SUE1) earnings surprise portfolios for which the daily probabilities are greater than or equal to 0.9 ($\pi_g \geq 0.9$ and $\pi_b \geq 0.9$), as well as the difference between the average abnormal buying and selling probabilities for the two portfolios. The table shows that 22.6% of

¹⁹We also compute *SUE* using the median value of EPSs forecasted by analysts: the patterns are qualitatively similar. However, this reduces the sample size substantially. We also conduct analyses equivalent to those reported in Table 8 using the DY-based measures and find similar patterns for $\pi_g^{DY}(-5, -1)$ and $\pi_b^{DY}(-5, -1)$.

firms in SUE10 have informed buying with high probability on day $d-1$, compared with only 16.5% of firms in SUE1. The difference extends back to day $d-5$, when 15.2% of firms in SUE10 have informed buying with high probability, compared with 13.0% of the firms in SUE1. This is consistent with informed trading on good news up to five days before the EAD. For informed selling, we find the opposite pattern: the fraction of firms for which π_b is large ($\pi_b \geq 0.9$) is always greater in SUE1 than in SUE10. The difference in the fraction between SUE1 and SUE10 is largest (6.5%) on day $d-1$ and is higher before the announcement date than after.

The relation between informed trading and the magnitude of the earnings surprise is also apparent in the row labeled $(\pi_g^{abn} - \pi_b^{abn})$, which presents the difference between the two abnormal probabilities. The table shows that, for example, the difference is 5.0% (10.8%) on day $d-5$ ($d-1$) for the high earnings-surprise portfolio, SUE10, as compared with -0.2% (-0.5%) for SUE1. This suggests that pre-EAD informed buying is prevalent in the high-surprise firms. It is also consistent with the existence of informed selling on bad news prior to the EAD.

As a robustness check, we consider an analysis equivalent to Table 9, where *SUE* is computed based on the median value of EPSs forecasted by analysts (i.e., instead of EPS_{it-4} , forecasted EPSs are used for Ψ in the definition above). Although the average number of component stocks in each portfolio decreases substantially, we find that the pattern is qualitatively similar. These results are available in the Internet Appendix.

4.3. Pre-EAD Informed Trading and Attenuation of Announcement Returns

We have already noted that (the probability of) informed buying before unscheduled good-news announcements, merger bids and dividend initiations, and informed selling before bad-news announcements, SEOs, attenuates the price response to the public announcements. For earnings announcements, we know not only whether it is good or

bad news but also the salience of the news as measured by SUE , and can therefore test whether pre-EAD informed trading also attenuates the price response to the salience of subsequent earnings news. To test this hypothesis, we estimate the following regressions separately for positive and negative earnings surprises (SUE):

$$CAR(0, +1) = a^+ + b^+ SUE + c^+ \pi_g(-5, -1) + \sum_{n=1}^N h_n X_n + e, \text{ for } SUE > 0 \quad (5)$$

$$CAR(0, +1) = a^- + b^- SUE + c^- \pi_b(-5, -1) + \sum_{n=1}^N h_n X_n + \varepsilon, \text{ for } SUE < 0, \quad (6)$$

where X_n ($n = 1, 2, \dots, N$) are the averages of the control variables over days $d-5$ to $d-1$. If prices respond to the magnitude of the earnings surprise, the coefficients on SUE , b^+ and b^- , are expected to be positive. Furthermore, the attenuation hypothesis predicts that c^+ is negative and c^- is positive.

Specifications (i) and (iii) in Panels A and B of Table 10 report the results of Fama and MacBeth (1973) cross-sectional estimates of the above equations, run quarterly from the second quarter of 1983 to the fourth quarter of 2013 (123 quarters). There are about 650 (410) positive (negative) earnings surprises on average each quarter. The two regressions show that the coefficient on SUE is positive and significant for both positive and negative earnings surprises. More importantly, the estimate of c^+ (c^-) is negative (positive) and significant, consistent with the attenuation hypothesis. When $CAR(-1, +1)$ is used as the dependent variable in Panels D and E the results are robust.

Columns (ii) and (iv) in Panels A and B of Table 10 also report the results of regressions, in which the earnings surprise (SUE) is interacted with the probabilities of

pre-EAD informed trading:

$$CAR(0, +1) = a^+ + b^+ SUE + c^+ \pi_g(-5, -1) + d^+ SUE^* \pi_g(-5, -1) + \sum_{n=1}^N h_n X_n + e,$$

for $SUE > 0$

(7)

$$CAR(0, +1) = a^- + b^- SUE + c^- \pi_b(-5, -1) + d^- SUE^* \pi_b(-5, -1) + \sum_{n=1}^N h_n X_n + \varepsilon,$$

for $SUE < 0$

(8)

The estimates of d^+ and d^- in regressions (ii) and (iv) in Table 10 are negative and significant for both positive and negative earnings surprises (SUE), so that pre-EAD informed trading does reduce the sensitivity of the price response to the salience of the earnings news.

Finally, column (v) in Panel C reports estimates of a single equation, which combines positive and negative earnings surprises:

$$CAR(0, +1) = a + \theta SUE + \gamma \pi_i(-5, -1) + \phi SUE^* \pi_i(-5, -1) + \sum_{n=1}^N h_n X_n + \omega D_{SUE} + \epsilon,$$

$i = g$ if $SUE > 0$ and $i = b$ if $SUE < 0$,

(9)

where D_{SUE} is a dummy variable that is equal to 1 if $SUE < 0$ and 0 otherwise. Panel C shows that the coefficient (θ) on SUE is positive and significant, and the coefficient (ϕ) on the interaction term ($SUE^* \pi_i$) is negative and significant. In all three regressions with an interaction term [specifications (ii), (iv), and (v)], the coefficient on $SUE^* \pi_i(-5, -1)$ is larger by a factor of 2-5 than the coefficient on SUE . This implies that an average daily probability of about 0.20-0.50 on the days preceding the earnings announcement

would eliminate entirely the price response to the announcement. When $CAR(-1, +1)$ is used in Eqs. (7)-(9), Panels D-F show that the results are qualitatively similar.

These results provide evidence in support both of the attenuation hypothesis and of the ability of the variables π_g and π_b to capture the intensity of informed trading. They confirm the findings of Vega (2006), who finds that higher values of PIN are associated with lower post-earnings-announcement drift. The main difference between her study and ours is that her independent variable is the unconditional PIN estimated in a period of at least 40 trading days (i.e., only *one* estimate) before the EAD rather than the posterior probabilities (estimated *each* day around the EAD) that we use, and she does not distinguish between informed trading on good news and on bad news. Possibly as a consequence, she finds only modest statistical significance for her estimates.

4.4. Post-EAD Informed Trading and Returns

Our goal here is to explore if the probabilities estimated in the post-EAD period do in fact capture informed trading. Similarly to the post-event analyses in Tables 6 and 7 for the three unscheduled corporate events (M&As, dividend initiations, and SEOs), we examine the relation between the informed-trading probabilities measured immediately after the EAD and subsequent stock returns by running quarterly Fama-MacBeth (1973) regressions, in which the dependent variable is the post-EAD return over 13 trading days, $CAR(+3, +15)$, and the explanatory variables are the probabilities averaged over two trading days ($d+1$ and $d+2$) immediately after the earnings announcement, $\pi_g(+1, +2)$ and $\pi_b(+1, +2)$.²⁰ We include the cumulative abnormal return, $CAR(+1, +2)$, as well as the analogs of the other six variables as controls.

The results are reported in Table 11. In both specifications, the probability of in-

²⁰We use two days after the announcement, because Figure 6 indicates that the probabilities of informed buying and selling are substantially higher on those days ($d+1$ and $d+2$) relative to other days, but experimenting with other windows leads to substantively similar results.

formed buying, $\pi_g(+1, +2)$, is positively and significantly associated with $CAR(+3, +15)$. In terms of economic significance, a one-standard-deviation move in $\pi_g(+1, +2)$ increases $CAR(+3, +15)$ by 0.2%, and this magnitude is comparable to that reported in Table 3. The probability of informed selling, $\pi_b(+1, +2)$, is also negatively associated with subsequent returns, although the coefficient on this variable in specification (ii) is smaller (in absolute terms) than that on the informed buying variable. Nevertheless, the results leave no doubt that there is informed trading after quarterly earnings announcements.²¹

5. Conclusion

Since the trades of privately informed agents improve the efficiency of financial markets, it is important to understand the activities of informed traders. In this paper, we use a comprehensive set of transactions data to analyze informed trading before and, importantly, *after* scheduled and unscheduled corporate announcements, and also examine the links between our informed-trading measures and stock returns around and after the announcement. While there are many models that suggest informed trading prior to an announcement, Kim and Verrecchia (1994), among others, show that informed trading may also occur after the announcement, making our post-announcement analyses of informed trading around a variety of corporate events particularly pertinent.

Many studies of informed trading have relied on estimates of EKOP's (1996) *PIN* calculated using long windows of daily data. These long estimation periods limit the precision and reliability of the conclusions, particularly when the focus is on trading around corporate events, since informed trading is likely to be concentrated on only a few

²¹The results using $CAR(+3, +20)$ as the dependent variable (reported in the Internet Appendix) are robust. We also run the analyses equivalent to Table 10 using the DY-based measures: $\pi_g^{DY}(-5, -1)$ and $\pi_b^{DY}(-5, -1)$, together with $\pi_{ls}^{DY}(-5, -1)$ as a control variable. But the result is not consistent with the attenuation hypothesis. In addition, we control for limit-order imbalance, $LOIMB(+1, +2)$, instead of $OIMB(+1, +2)$ in Table 11. The results are consistent with those in Table 11.

days around the announcement. We develop daily conditional probabilities of informed buying and informed selling that incorporate the information on each day's buy and sell orders, in the spirit of the EKOP (1996) model.

Using data on M&As, dividend initiations, and SEOs, we find evidence of informed trading both before and after these unscheduled corporate announcements. Our analyses reveal that a significant part of the information in the three types of announcements is impounded in the share price by informed trading before the announcements. The conditional daily probabilities of informed trading after merger bids contain information about future bid withdrawals, the emergence of competing bidders, and subsequent stock returns to target shareholders, and we also find evidence of informed trading following announcements of dividend initiations and SEOs.

Similarly, our analysis of the conditional probabilities around quarterly earnings announcements yields evidence of informed trading both before and after the announcements. High probabilities of informed trading prior to the announcement reduce the stock price response to earnings surprises, suggesting that the informed trading that we identify impounds information about the forthcoming earnings news into the stock price before the announcement. The conditional probabilities after the earnings announcement contain information about future stock returns, confirming that these probabilities are also related to informed trading.

While we acknowledge that the ability of *PIN* to identify informed trading has been disputed in the literature, the totality of our evidence accords with the ability of the model to identify occasions of informed trading before and after public announcements, and to identify informed trading that is predictive of subsequent events and security returns. Our evidence of post-announcement informed trading offers support for frameworks such as Kim and Verrecchia (1994, 1997) in which different agents have different interpretations of public information signals, or models in which different agents have different abilities

to process the information in public announcements. The fact that we are able to identify informed trading following public announcements points to caution in interpreting trading on information as trading on private information.

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Table 1
Descriptive Statistics and Distributions of the Probabilities of Informed Trading and Other Variables for NYSE/AMEX-Listed Stocks

Descriptive Statistics of NYSE/AMEX Stocks for the Sample Period (7,626 Trading Days: 1983:04-2013:12)									
Period	Variables	Mean	Median	Max	Min	STD	CV	Skewness	Kurtosis
		Panel A: Daily Number of Quote Updates and Trades							
Whole Period	N_QuoteUpdate	19,843.32	7,852.65	417,245.88	2.92	34,142.39	154.45	4.28	35.07
	N_Trade	2,120.76	763.09	71,069.70	1.12	4,318.05	212.11	7.74	112.08
Non-HFT Era	N_QuoteUpdate	1,389.54	690.02	44,939.05	1.34	2,774.35	150.74	4.27	36.13
	N_Trade	374.41	149.70	16,715.05	1.00	854.78	217.18	8.18	124.08
HFT Era	N_QuoteUpdate	81,443.57	31,762.13	1,660,037.50	8.18	138,851.54	166.83	4.29	31.53
	N_Trade	7,950.21	2,810.63	252,510.12	1.53	15,878.73	195.20	6.25	72.05
Panel B: Daily Posterior Probabilities									
Whole Period	$\pi_{\emptyset}$	0.679	0.910	1.000	0.000	0.418	64.59	-0.87	-0.68
	π_g	0.197	0.008	1.000	0.000	0.356	203.15	1.73	1.83
	π_b	0.124	0.001	1.000	0.000	0.291	265.86	2.61	6.49
Non-HFT Era	$\pi_{\emptyset}$	0.672	0.900	1.000	0.000	0.414	64.69	-0.83	-0.75
	π_g	0.212	0.010	1.000	0.000	0.362	188.24	1.59	1.33
	π_b	0.117	0.001	1.000	0.000	0.278	269.08	2.73	7.31
HFT Era	$\pi_{\emptyset}$	0.704	0.941	1.000	0.000	0.430	64.28	-1.01	-0.47
	π_g	0.147	0.000	1.000	0.000	0.336	252.81	2.19	3.51
	π_b	0.149	0.001	1.000	0.000	0.335	255.14	2.21	3.74

(Table 1: continued)

Descriptive Statistics of NYSE/AMEX Stocks for the Sample Period (7,626 Trading Days: 1983:04-2013:12)									
Period	Variables	Mean	Median	Max	Min	STD	CV	Skewness	Kurtosis
Panel C: DY-based Daily Probabilities									
Whole Period	π_g^{DY}	0.143	0.003	1.000	0.000	0.291	215.33	2.18	3.69
	π_b^{DY}	0.131	0.002	1.000	0.000	0.279	226.54	2.37	4.70
	π_{ls}^{DY}	0.195	0.008	1.000	0.000	0.356	199.51	1.69	1.47
Non-HFT Era	π_g^{DY}	0.148	0.004	1.000	0.000	0.289	206.38	2.12	3.52
	π_b^{DY}	0.132	0.003	1.000	0.000	0.274	220.57	2.36	4.79
	π_{ls}^{DY}	0.211	0.010	1.000	0.000	0.364	185.56	1.53	0.89
HFT Era	π_g^{DY}	0.126	0.000	1.000	0.000	0.297	245.14	2.37	4.24
	π_b^{DY}	0.126	0.000	1.000	0.000	0.296	246.45	2.39	4.42
	π_{ls}^{DY}	0.142	0.000	1.000	0.000	0.326	245.95	2.20	3.40
Panel D: Other Key Variables (Daily)									
Whole Period	R	0.001	0.000	0.469	-0.312	0.034	625.38	1.99	80.23
	PSPR	1.539	0.695	142.655	-0.654	5.197	256.66	8.62	177.37
	OIMB	10.123	11.807	99.954	-99.845	35.939	547.09	-0.39	2.62
	TURN	0.005	0.003	0.187	0.000	0.009	172.10	9.65	188.29
	SIZE	13.131	13.212	19.046	7.068	2.027	15.45	-0.09	-0.29
	BTM	0.718	0.578	41.349	0.006	1.419	109.18	6.39	116.09
Non-HFT Era	R	0.001	0.000	0.481	-0.326	0.035	704.11	1.93	85.88
	PSPR	1.847	0.869	178.813	-0.854	6.358	256.81	9.00	197.41
	OIMB	13.322	15.360	100.000	-99.829	40.593	421.84	-0.42	1.41
	TURN	0.004	0.002	0.140	0.000	0.006	177.68	9.66	186.27
	SIZE	12.898	12.949	18.824	6.970	2.017	15.63	-0.04	-0.36
	BTM	0.698	0.587	10.848	0.007	0.610	85.79	5.85	91.34
HFT Era	R	0.001	0.000	0.426	-0.262	0.033	363.18	2.17	61.42
	PSPR	0.516	0.116	22.334	0.012	1.332	256.15	7.36	110.71
	OIMB	-0.531	-0.028	99.803	-99.897	20.440	964.25	-0.26	6.63
	TURN	0.010	0.007	0.343	0.000	0.016	153.52	9.63	195.01
	SIZE	13.907	14.088	19.786	7.393	2.059	14.82	-0.24	-0.08
	BTM	0.784	0.548	142.933	0.003	4.115	187.08	8.22	198.53
Panel E: Monthly Unconditional Probabilities (369 Months: 1983:04-2013:12)									
Whole Period	$(1 - \alpha)$	0.689	0.695	0.999	0.004	0.163	23.65	-0.58	1.55
	$\alpha(1 - \delta)$	0.186	0.168	0.936	0.000	0.132	71.33	1.13	2.52
	$\alpha\delta$	0.125	0.091	0.905	0.000	0.126	105.33	1.91	5.73
Non-HFT Era	$(1 - \alpha)$	0.692	0.703	0.999	0.000	0.170	24.71	-0.74	1.60
	$\alpha(1 - \delta)$	0.195	0.176	0.941	0.000	0.136	70.42	1.12	2.43
	$\alpha\delta$	0.113	0.075	0.902	0.000	0.127	114.34	2.16	6.79
HFT Era	$(1 - \alpha)$	0.677	0.668	1.000	0.016	0.136	20.05	-0.05	1.41
	$\alpha(1 - \delta)$	0.158	0.139	0.919	0.000	0.117	74.42	1.16	2.84
	$\alpha\delta$	0.165	0.144	0.916	0.000	0.123	74.76	1.06	2.14

Panel F: Distribution of the Daily Conditional Probabilities for All Firm-Days										
	0.00-0.10	0.10-0.20	0.20-0.30	0.30-0.40	0.40-0.50	0.50-0.60	0.60-0.70	0.70-0.80	0.80-0.90	0.90-1.00
π_ϕ	0.261	0.018	0.013	0.012	0.012	0.013	0.015	0.019	0.032	0.607
π_g	0.748	0.024	0.014	0.010	0.009	0.008	0.008	0.009	0.012	0.151
π_b	0.843	0.017	0.010	0.008	0.006	0.006	0.006	0.006	0.008	0.090

This table reports descriptive statistics of the numbers of daily quote updates and transactions (Panel A), daily posterior (or conditional) probabilities (Panel B), Duarte and Young (2009) (DY)-based daily probabilities (Panel C), other key variables (Panel D), and monthly unconditional (or prior) probabilities (Panel E), as well as the distribution of the posterior daily probabilities for all firm-days (Panel F). Panels A-E report the descriptive statistics

for the two subperiods: non-high-frequency-trading era (non-HFT era: 1983-2006), during which the Lee-Ready (1991) algorithm is used to classify trades; and HFT era (2007-2013), during which the Holden-Jacobsen (2014) algorithm is used. The cross-sectional value for each statistic is calculated each day (Panels A-D) or each month (Panel E) and then the time-series average of those values is reported. The variables are defined as follows. *N_QuoteUpdate*: the number of daily quote updates, where a quote update is defined as any change in the prevailing intraday best bid or offer price (BBO), or any change in the displayed size (depth) for the BBO across all exchanges, following Conrad, Wahal, and Xiang (2015); *N_Trade*: the number of trades (transactions) executed across all exchanges within each day; π_0 : the estimated posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that no information event occurs on a given trading day; π_g : the estimated posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that a good-news information event occurs on a given day; π_b : the estimated posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that a bad-news information event occurs on a given day; π_g^{DY} : the DY-based daily probability of trading on good-news information; π_b^{DY} : the DY-based daily probability of trading on bad-news information; π_{ls}^{DY} : the DY-based daily probability related to a liquidity shock (unrelated to information asymmetry); $(1 - \alpha)$: the monthly estimated unconditional probability that no information event occurs (hence α is the probability with which an information event occurs) on a day; $\alpha(1 - \delta)$: the monthly estimated unconditional probability that a good news information event occurs on a day (δ is the probability with which the information event contains bad news); $\alpha\delta$: the monthly estimated unconditional probability that a bad news information event occurs on a day; *R*: the daily stock return; *PSPR*: the daily proportional quoted spread, which is the average of intraday proportional quoted spreads in % [i.e., ((dollar spread)/(quote mid-point))*100]; *OIMB*: the daily market-order imbalance (in %) [i.e., ((#BUY - #SELL)/(#BUY + #SELL))*100]; *TURN*: daily share turnover; *SIZE*: the natural logarithm of daily market value (in \$million); *BTM*: the book-to-market ratio (quarter-end book equity divided by market value of equity). For estimating the monthly unconditional probabilities and daily conditional probabilities, each trade in the ISSM/TAQ databases is classified as buyer-initiated or seller-initiated via the Lee-Ready (1991) algorithm up to 2006 and the Holden-Jacobsen (2014) algorithm for the 2007-2013 period. The sample periods for NYSE/AMEX stocks are the past 7,626 trading days from April 1983 to December 2013 (Panels A, B, C, and D), or the past 369 months from April 1983 to December 2013 (Panel E). The sample includes common stocks only (SHRCD = 10 or 11 in CRSP). The average number of component stocks used each day (or month) is 1,684.0-2,132.8 for the whole sample period, 1,725.3-1,914.5 for the non-HFT era, and 1,266.8-3,140.2 for the HFT era, depending on the availability of each variable.

Table 2

Correlations between Daily Posterior Probabilities, DY-Based Daily Probabilities, Monthly Unconditional (Prior) Probabilities of Informed Trading, and Other Variables

Panel A: Time-Series Average of Cross-Sectional Correlations of Daily Posterior Probabilities and Other Variables												
Variables	$\pi_{\emptyset}$	π_g	π_b	π_g^{DY}	π_b^{DY}	π_{ls}^{DY}	R	PSPR	OIMB	TURN	SIZE	BTM
$\pi_{\emptyset}$	1											
π_g	-0.721	1										
π_b	-0.535	-0.180	1									
π_g^{DY}	-0.344	0.504	-0.107	1								
π_b^{DY}	-0.193	-0.133	0.445	-0.149	1							
π_{ls}^{DY}	-0.474	0.349	0.269	0.033	-0.025	1						
R	-0.083	0.160	-0.080	0.163	-0.113	0.056	1					
PSPR	0.090	-0.071	-0.031	-0.040	-0.040	-0.003	-0.004	1				
OIMB	-0.057	0.265	-0.247	0.285	-0.301	0.001	0.233	-0.047	1			
TURN	-0.250	0.219	0.086	0.133	0.058	0.167	0.057	-0.085	0.037	1		
SIZE	-0.142	0.106	0.056	0.055	0.053	-0.014	0.008	-0.541	0.053	0.102	1	
BTM	0.034	-0.036	0.002	-0.015	-0.008	0.004	0.004	0.211	-0.043	-0.029	-0.304	1

Panel B: Joint Distribution of Extreme Values of Daily Posterior Probabilities		
	$\pi_g < 0.1$	$\pi_g > 0.9$
$\pi_b < 0.1$	0.611	0.158
$\pi_b > 0.9$	0.090	0.000

Panel C: Time-Series Average of Cross-Sectional Correlations of Monthly Unconditional Probabilities			
Measures	$(1 - \alpha)$	$\alpha(1 - \delta)$	$\alpha\delta$
$(1 - \alpha)$	1		
$\alpha(1 - \delta)$	-0.647	1	
$\alpha\delta$	-0.596	-0.214	1

This table reports averages of correlations between the daily posterior probabilities of informed trading, Duarte and Young (2009) (DY)-based daily probabilities, and other key variables, as well as those between the monthly unconditional (prior) probabilities. Panel A reports time-series averages of the daily cross-sectional correlations between the posterior probabilities, DY-based probabilities, and other variables. Panel B reports the joint distribution of firm-days for which the two daily posterior probabilities are extreme: i.e., (i) both smaller than 0.1; (ii) both larger than 0.9; and (iii) one smaller than 0.1 and the other larger than 0.9. Panel C reports time-series averages of the cross-sectional correlations between the unconditional probabilities of informed trading that are estimated each month. The variables are defined as follows. π_0 : the estimated posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that no information event occurs on a given trading day; π_g : the estimated posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that a good-news information event occurs on a given day; and π_b : the estimated posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that a bad-news information event occurs on a given day; π_g^{DY} : the DY-based daily probability of trading on good-news information; π_b^{DY} : the DY-based daily probability of trading on bad-news information; π_{ls}^{DY} : the DY-based daily probability related to a liquidity shock (unrelated to information asymmetry); $(1 - \alpha)$: the unconditional probability that no information event occurs; $\alpha(1 - \delta)$: the unconditional probability that a good news information event occurs on a given day (δ is the probability with which the information event contains bad news); $\alpha\delta$: the unconditional probability that a bad news information event occurs on a day; R: the daily stock return; PSPR: the daily proportional quoted spread, which is the average of intradaily proportional quoted spreads in % [i.e., ((dollar spread)/(quote midpoint))*100]; OIMB: the daily market-order imbalance (in %) [i.e., ((#BUY - #SELL)/(#BUY + #SELL))*100]; TURN: daily share turnover; SIZE: the natural logarithm of daily market value (in \$million); BTM: the book-to-market ratio (quarter-end book equity divided by market value of equity). For estimating the monthly unconditional probabilities and daily posterior probabilities, each trade in the ISSM/TAQ databases is classified as buyer-initiated or seller-initiated via the Lee-Ready (1991) algorithm up to 2006 and the Holden-Jacobsen (2014) algorithm for the 2007-2013 period. The sample period for NYSE/AMEX stocks is the past 7,626 trading days from April 1983 to December 2013 in Panels A and B, while it is the past 369 months from April 1983 to December 2013 in Panel C. The sample includes common stocks only (SHRCD = 10 or 11 in CRSP). The average number of component stocks used each day (or month) is 1,684.0 to 1,876.8, depending on the availability of each variable.

Table 3
Pre-Announcement Informed Trading in Three Unscheduled Corporate Events and the Announcement Return

Panel A1: Pre-Announcement Informed Trading around M&AD and the Announcement Return for Target Firms												
Explana. Var.	Dep. Var. = $CAR(0, +1)$											
	(i)		(ii)		(iii)		(iv)		(v)		(vi)	
Intercept	0.236	***	0.294	***	0.193	***	0.298	***	0.235	***	0.299	***
	27.47		9.30		27.53		9.53		24.24		9.58	
$\pi_g(-20, -1)$	-0.166	***	-0.104	***					-0.156	***	-0.090	***
	-6.50		-3.53						-6.25		-3.13	
$\pi_b(-20, -1)$					-0.020		-0.050		-0.036		-0.050	
					-0.56		-1.33		-1.02		-1.32	
R(-20, -1)			-3.431	***			-3.632	***			-3.269	***
			-4.71				-5.10				-4.54	
RVOLA(-20, -1)			2.092	***			1.722	***			1.789	***
			6.76				5.58				5.80	
PSPR(-20, -1)			-0.001				-0.001				-0.001	
			-0.45				-0.40				-0.41	
OIMB(-20, -1)			-0.001	**			-0.001	***			-0.001	**
			-2.25				-3.97				-2.44	
TURN(-20, -1)			-1.979	***			-2.005	***			-1.764	**
			-2.67				-2.76				-2.42	
SIZE(-20, -1)			-0.016	***			-0.017	***			-0.015	**
			-3.83				-4.30				-3.78	
BTM			0.005				0.007				0.006	
			0.48				0.65				0.57	
R ²	0.016		0.053		0.000		0.043		0.016		0.047	
Obs	2527		2340		2498		2317		2496		2316	

(Table 3: continued)

Panel A2: Pre-Announcement Informed Trading around M&AD and the Announcement Return for Target Firms									
Explana. Var.	Dep. Var. = $CAR(-1, +1)$								
	(i)		(ii)		(iii)		(iv)		(v) (vi)
Intercept	0.240 ***		0.287 ***		0.200 ***		0.291 ***		0.239 *** 0.292 ***
	28.19		9.16		28.71		9.36		24.83 9.42
$\pi_g(-20, -1)$	-0.156 ***		-0.110 ***						-0.147 *** -0.097 ***
	-6.15		-3.78						-5.90 -3.40
$\pi_b(-20, -1)$					-0.021		-0.051		-0.036 -0.051
					-0.58		-1.35		-1.01 -1.35
R(-20, -1)			-2.479 ***				-2.690 ***		-2.301 ***
			-3.42				-3.80		-3.22
RVOLA(-20, -1)			2.377 ***				2.027 ***		2.096 ***
			7.78				6.65		6.87
PSPR(-20, -1)			-0.001				-0.001		-0.001
			-0.53				-0.48		-0.50
OIMB(-20, -1)			-0.001 **				-0.001 ***		-0.001 **
			-2.12				-3.95		-2.33
TURN(-20, -1)			-2.285 ***				-2.355 ***		-2.099 ***
			-3.12				-3.28		-2.92
SIZE(-20, -1)			-0.015 ***				-0.017 ***		-0.014 ***
			-3.62				-4.11		-3.55
BTM			0.004				0.006		0.005
			0.35				0.51		0.42
R ²	0.015		0.056		0.000		0.044		0.014 0.049
Obs	2527		2340		2498		2317		2496 2316

(Table 3: continued)

Panel B: Pre-Declaration Informed Trading around Dividend Initiation Date (DID) and the Declaration Return												
Explana. Var.	Dep. Var. = CAR(0, +1)											
	(i)		(ii)		(iii)		(iv)		(v)		(vi)	
Intercept	0.015	***	0.022		0.007	***	0.017		0.013	***	0.022	
	5.34		1.42		2.93		1.13		3.92		1.39	
$\pi_g(-20, -1)$	-0.027	***	-0.022	*					-0.027	***	-0.023	*
	-2.57		-1.72						-2.59		-1.76	
$\pi_b(-20, -1)$					0.021	*	0.000		0.021	*	0.001	
					1.72		0.02		1.73		0.10	
R(-20, -1)			-0.493				-0.561				-0.480	
			-1.26				-1.44				-1.22	
RVOLA(-20, -1)			-0.217				-0.279				-0.223	
			-0.97				-1.25				-0.99	
PSPR(-20, -1)			0.000				0.002				0.000	
			0.13				0.50				0.14	
OIMB(-20, -1)			0.000				0.000				0.000	
			0.21				-0.40				0.25	
TURN(-20, -1)			0.310				0.270				0.334	
			0.62				0.54				0.66	
SIZE(-20, -1)			-0.001				-0.001				-0.001	
			-0.75				-0.62				-0.73	
BTM			0.003				0.003				0.003	
			0.56				0.54				0.55	
R ²	0.009		0.021		0.004		0.015		0.014		0.021	
Obs	692		463		691		462		691		462	

(Table 3: continued)

Panel C: Pre-Announcement Informed Trading around SEO File Date (SEOD) and the Announcement Return												
Explana. Var.	Dep. Var. = CAR(0, +1)											
	(i)		(ii)		(iii)		(iv)		(v)		(vi)	
Intercept	-0.012	***	-0.019	***	-0.015	***	-0.021	***	-0.015	***	-0.021	***
	-6.85		-2.70		-9.58		-2.95		-7.16		-3.00	
$\pi_g(-20, -1)$	0.001		-0.005						0.001		-0.005	
	0.11		-0.82						0.23		-0.90	
$\pi_b(-20, -1)$					0.019	**	0.024	***	0.020	**	0.024	***
					2.54		2.87		2.55		2.89	
R(-20, -1)			0.070				0.058				0.079	
			0.43				0.36				0.48	
RVOLA(-20, -1)			0.039				0.020				0.032	
			0.41				0.21				0.34	
PSPR(-20, -1)			0.000				0.000				0.000	
			0.28				0.30				0.31	
OIMB(-20, -1)			0.000				0.000				0.000	*
			0.86				1.46				1.68	
TURN(-20, -1)			-0.046				-0.057				-0.040	
			-0.29				-0.36				-0.25	
SIZE(-20, -1)			0.001				0.000				0.000	
			0.76				0.28				0.43	
BTM			0.003	**			0.003	**			0.003	**
			2.09				2.12				2.13	
R ²	0.000		0.004		0.004		0.008		0.004		0.009	
Obs	1714		1610		1713		1609		1713		1609	

This table reports the results of regressions of the announcement return for three types of unscheduled corporate events on the pre-event average daily probabilities of informed trading for NYSE/AMEX-listed firms. The sample period is from 1983 to 2013. Panels A1 and A2 contain the analysis around the M&A announcement date (M&AD) for target firms in completed and withdrawn M&As, Panel B around the dividend initiation date (DID), and Panel C around the seasoned equity offering (SEO) file date (SEOD). A dividend declaration is considered as a dividend initiation date (DID) if it is the first (cash) dividend ever declared and paid in a firm's history (ordinary quarterly dividend only: i.e., distribution code = 1234 in Compustat). To be included in the SEO sample, filed amount should be at least \$25 million and at least some primary shares should be offered. The announcement return used as the dependent variable is measured by $CAR(0, +1)$ or $CAR(-1, +1)$, which is the cumulative abnormal return over two trading days around the event date (M&AD, DID, and SEOD: day d). Other variables are defined as follows. $\pi_g(-20, -1)$: the average of daily π_g 's from day -20 to day -1 (relative to each of the three event dates), where π_g is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is good news; $\pi_b(-20, -1)$: the average of daily π_b 's from day -20 to day -1, where π_b is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is bad news; $R(-20, -1)$: the average of daily stock returns from day -20 to day -1; $RVOLA(-20, -1)$: the return volatility, which is the standard deviation of daily returns from day -20 to day -1; $PSPR(-20, -1)$: the average of PSPR from day -20 to day -1, where PSPR is the daily mean of intradaily proportional quoted spreads in % [i.e., ((dollar spread)/(quote mid-point))*100]; $OIMB(-20, -1)$: the average of OIMB from day -20 to day -1, where OIMB is the daily order imbalance in % [i.e., ((#BUY - #SELL)/(#BUY + #SELL))*100]; $TURN(-20, -1)$: the average of daily share turnover from day -20 to day -1; $SIZE(-20, -1)$: the natural logarithm of the average MV from day +1 to day +10, where MV is the daily market value (in \$million); and BTM : the book-to-market ratio in the previous quarter. The values in the first row for each explanatory variable are the coefficients from the regressions, and the values italicized in the second row of each variable are t -statistics. R^2 is the R-squared from the regressions. Obs is the number of companies used in the regressions. Coefficients significantly different from zero at the significance levels of 1%, 5%, and 10% are indicated by ***, **, and *, respectively.

Table 4
Post-Announcement Informed Trading and Bid Withdrawals

Probit Regressions after the M&AD: Dep. Var. = 1 for Withdrawn Bids and 0 Otherwise				
Explana. Var.	Panel A: With All M&A Targets		Panel B: With All-Cash M&A Targets	
	(i)	(ii)	(iii)	(iv)
Intercept	-0.633 *** 0.000	-0.235 *** 0.143	-0.612 *** 0.000	0.199 *** 0.483
$\pi_g(+1, +10)$	-0.659 * 0.094	-0.396 0.480	-1.127 * 0.085	-1.146 0.243
$\pi_b(+1, +10)$	0.097 0.770	-0.081 0.823	1.485 *** 0.008	1.515 ** 0.016
$R(+1, +10)$		-0.065 0.780		-0.285 0.465
$RVOLA(+1, +10)$		-0.011 0.901		0.078 0.595
$PSPR(+1, +10)$		0.005 0.116		0.004 0.146
$OIMB(+1, +10)$		0.009 *** 0.000		0.014 *** 0.000
$TURN(+1, +10)$		-5.190 *** 0.000		-6.569 *** 0.000
$SIZE(+1, +10)$		-0.053 ** 0.012		-0.077 ** 0.043
BTM		0.164 *** 0.003		-0.043 0.646
Pseudo R ²	0.002	0.045	0.007	0.092
Obs	2613	2409	1008	930

This table reports the results of probit regressions of a bid withdrawal dummy variable on the averages of the daily posterior probabilities over the days following the initial bid announcement. The sample includes all target firms that are listed on the NYSE/AMEX over the 1983-2013 sample period. Panel A include all target firms, while Panel B includes only target firms that receive all-cash payment bids. The dependent variable is equal to 1 for M&A bids that are withdrawn and 0 for M&A bids that are completed. The explanatory variables are defined as follows. $\pi_g(+1, +10)$: the average of daily π_g 's from day +1 to day +10 (relative to the initial M&A announcement date, M&AD), where π_g is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is good news; $\pi_b(+1, +10)$: the average of daily π_b 's from day +1 to day +10 (relative to the M&AD), where π_b is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is bad news; $R(+1, +10)$: the average of daily stock returns from day +1 to day +10 (relative to the M&AD); $RVOLA(+1, +10)$: the return volatility, which is the standard deviation of daily returns from day +1 to day +10; $PSPR(+1, +10)$: the average of PSPR from day +1 to day +10, where PSPR is the daily mean of intradaily proportional quoted spreads in % [i.e., ((dollar spread)/(quote mid-point))*100]; $OIMB(+1, +10)$: the average of OIMB from day +1 to day +10, where OIMB is the daily order imbalance in % [i.e., ((#BUY - #SELL)/(#BUY + #SELL))*100]; $TURN(+1, +10)$: the average of daily share turnover from day +1 to day +10; $SIZE(+1, +10)$: the natural logarithm of the average MV from day +1 to day +10, where MV is the daily market value (in \$million); and BTM : the book-to-market ratio in the previous quarter. The values in the first row for each explanatory variable are the coefficients from the regressions, and the values italicized in the second row of each variable are *p*-values. Pseudo R² is the pseudo R-squared from the probit regressions. Obs is the number of target firms used in each of the regressions. Coefficients significantly different from zero at the significance levels of 1%, 5%, and 10% are indicated by ***, **, and *, respectively.

Table 5
Post-Announcement Informed Trading and the Emergence of Competing Bids

Probit Regressions after the M&AD: Dep. Var. = 1 for Emergence of Competing Bidders and 0 Otherwise				
Explana. Var.	Panel A: With All M&A Targets		Panel B: With All-Cash M&A Targets	
	(i)	(ii)	(iii)	(iv)
Intercept	-1.221 *** 0.000	-2.019 *** 0.000	-1.408 *** 0.000	-2.009 *** 0.000
$\pi_g(+1, +10)$	1.114 ** 0.015	1.798 *** 0.006	1.736 ** 0.029	2.861 ** 0.015
$\pi_b(+1, +10)$	-0.907 ** 0.017	-1.022 ** 0.012	0.069 0.916	0.179 0.805
$R(+1, +10)$		-0.392 0.159		-1.092 ** 0.025
$RVOLA(+1, +10)$		-0.008 0.936		0.169 0.367
$PSPR(+1, +10)$		0.004 0.150		0.004 0.156
$OIMB(+1, +10)$		-0.002 0.213		0.000 0.840
$TURN(+1, +10)$		-1.996 0.118		-1.954 0.258
$SIZE(+1, +10)$		0.106 *** 0.000		0.081 * 0.087
BTM		0.182 *** 0.006		0.034 0.781
Pseudo R ²	0.002	0.014	0.010	0.021
Obs	2558	2357	964	889

This table reports the results of probit regressions of a multi-bidder dummy variable on the average the daily posterior probabilities over the days following the initial bid announcement. The sample includes all target firms that are listed on the NYSE/AMEX over the 1983-2013 sample period. Panel A include all target firms, while Panel B includes only target firms that receive all-cash payment bids. The dependent variable is equal to 1 if a target firm faces any competing bidder(s) after the initial bidder announces its M&A intention and 0 otherwise. The explanatory variables are defined as follows. $\pi_g(+1, +10)$: the average of daily π_g 's from day +1 to day +10 relative to the initial announcement date (M&AD), where π_g is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is good news; $\pi_b(+1, +10)$: the average of daily π_b 's from day +1 to day +10 relative to the M&AD, where π_b is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is bad news; $R(+1, +10)$: the average of daily stock returns from day +1 to day +10 relative to the M&AD; $RVOLA(+1, +10)$: the return volatility, which is the standard deviation of daily returns from day +1 to day +10; $PSPR(+1, +10)$: the average of PSPR from day +1 to day +10, where PSPR is the daily mean of intradaily proportional quoted spreads in % [i.e., ((dollar spread)/(quote mid-point))*100]; $OIMB(+1, +10)$: the average of OIMB from day +1 to day +10, where OIMB is the daily order imbalance in % [i.e., ((#BUY - #SELL)/(#BUY + #SELL))*100]; $TURN(+1, +10)$: the average of daily share turnover from day +1 to day +10; $SIZE(+1, +10)$: the natural logarithm of the average MV from day +1 to day +10, where MV is the daily market value (in \$million); and BTM : the book-to-market ratio in the previous quarter. The values in the first row for each explanatory variable are the coefficients from the regressions, and the values italicized in the second row of each variable are *p*-values. *Pseudo R*² is the pseudo R-squared from the probit regressions. *Obs* is the number of target firms used in each of the regressions. Coefficients significantly different from zero at the significance levels of 1%, 5%, and 10% are indicated by ***, **, and *, respectively.

Table 6
Predictive Regressions of the Return or the Return Difference After the M&A Announcement Date

Predictive Regressions of the Return or the Return Difference After the M&A Announcement Date (M&AD)						
Explana. Var.	Panel A1: For All Targets		Panel A2: For All-Cash Targets		Panel B: For All-Stock M&A Bidders and Targets	
	Dep. Var. = $R(+11, FND)$				Dep. Var. = $DR(+11, FND)$	
	(i)	(ii)	(ia)	(iia)	(iii)	(iv)
Intercept	0.016	-0.177	-0.057 **	-0.063	-0.068	0.397 **
	0.54	-1.57	-2.22	-0.59	-1.51	2.04
$\pi_g(+1, +10)$	-0.257	-0.129	0.648 ***	0.723 **	0.794 **	0.259
	-0.97	-0.35	2.62	2.14	2.01	0.49
$\pi_b(+1, +10)$	0.173	0.161	-0.193	-0.166	-0.272	-0.243
	0.78	0.68	-0.95	-0.80	-0.89	-0.71
$R(+1, +10)$		-0.140		-0.113		0.186
		-0.88		-0.78		0.90
$RVOLA(+1, +10)$		0.038		0.038		0.044
		0.65		0.68		0.53
$PSPR(+1, +10)$		-0.001		0.000		-0.138 ***
		-0.31		0.01		-3.02
$OIMB(+1, +10)$		-0.001		-0.001		0.000
		-1.63		-1.45		0.05
$TURN(+1, +10)$		-0.704		0.298		-2.373
		-1.12		0.74		-1.41
$SIZE(+1, +10)$		0.025 *		-0.002		-0.047 **
		1.68		-0.13		-2.10
BTM		0.020		-0.053		-0.058
		0.50		-1.43		-0.91
R^2	0.000	0.003	0.010	0.017	0.023	0.084
Obs	2372	2196	831	772	228	215

This table reports the results of predictive regressions after the M&A announcement date (M&AD) over the sample period from 1983 to 2013. Panels A1 and A2 report regressions of the M&A post-announcement return on the average daily conditional probabilities over the days after the M&AD for NYSE/AMEX-listed target firms. Panel A1 include all target firms, while Panel A2 includes only target firms that receive all-cash payment bids. The dependent variable in Panels A1 and A2, $R(+11, FND)$, is the average of daily returns from day +11 after the M&AD to the final date (FND), where FND is the

effective date for completed M&As and the withdrawn date for withdrawn M&As. Panel B reports the results of regressions of the average post-M&AD return difference between NYSE/AMEX-listed acquiring firms and target firms on the average daily conditional probabilities of target firms over the days after the M&AD. The sample in Panel B includes only target firms that receive all-stock payment bids. The dependent variable in Panel B is $DR(+11, FND)$, which is the average of daily return differences ($R_{Target} - R_{Bidder}$) from day +11 after the M&AD to the final date (FND). To calculate the daily return differences, each target firm is matched with its bidder firm both listed on the NYSE/AMEX. Other variables are defined as follows. $\pi_g(+1, +10)$: the average of daily π_g 's from each event's day +1 to day +10 (relative to the initial M&AD), where π_g is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is good news; $\pi_b(+1, +10)$: the average of daily π_b 's from each event's day +1 to day +10, where π_b is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is bad news; $R(+1, +10)$: the average of daily stock returns from each event's day +1 to day +10; $RVOLA(+1, +10)$: the return volatility, which is the standard deviation of daily returns from each event's day +1 to day +10; $PSPR(+1, +10)$: the average of PSPR from each event's day +1 to day +10, where PSPR is the daily mean of intradaily proportional quoted spreads in % [i.e., ((dollar spread)/(quote mid-point))*100]; $OIMB(+1, +10)$: the average of OIMB from each event's day +1 to day +10, where OIMB is the daily (market-)order imbalance in % [i.e., ((#BUY - #SELL)/(#BUY + #SELL))*100]; $TURN(+1, +10)$: the average of daily share turnover from each event's day +1 to day +10; $SIZE(+1, +10)$: the natural logarithm of the average MV from each event's day +1 to day +10, where MV is the daily market value (in \$million); and BTM : the book-to-market ratio in the previous quarter. The values in the first row for each explanatory variable are the coefficients from the regressions, and the values italicized in the second row of each variable are t -statistics. All coefficients are multiplied by 100. R^2 is the R-squared from the regressions. Obs is the number of target companies used in the regressions. Coefficients significantly different from zero at the significance levels of 1%, 5%, and 10% are indicated by ***, **, and *, respectively.

Table 7
Predictive Regressions After the Announcement Dates of Dividend Initiations and Seasoned Equity Offerings

Predictive Regressions After the Announcement Dates of Dividend Initiations and SEOs										
Explana. Var.	Dep. Var. = $R(+11, +60)$				Dep. Var. = $R(+11, +30)$					
	Panel A: After the DID		Panel B: After the SEOD		Panel C: After the DID		Panel D: After the SEOD			
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)		
Intercept	-0.009	0.094	0.032	**	0.056	0.005	0.017	0.042	0.193	*
	-0.46	1.08	1.98		0.85	0.19	0.12	1.63	1.87	
$\pi_g(+1, +10)$	0.158	***	0.121	**	-0.009	-0.032	0.176	**	0.158	*
	3.25		2.07		-0.25	-0.73	2.56		1.67	
$\pi_b(+1, +10)$	-0.034		-0.011		-0.105	**	-0.152	***	0.044	0.037
	-0.59		-0.17		-2.09		-2.75		0.54	0.35
$R(+1, +10)$			-2.256		-1.215		-0.836			
			-1.60		-1.32		-0.37			
$RVOLA(+1, +10)$			0.965		-0.465		0.978			
			1.32		-1.14		0.83			
$PSPR(+1, +10)$			-0.017		-0.042	***	-0.009			
			-0.89		-2.80		-0.30			
$OIMB(+1, +10)$			0.000		0.000		0.000			
			-0.32		-0.58		-0.23			
$TURN(+1, +10)$			-0.930		-0.983		-0.680			
			-0.46		-1.11		-0.21			
$SIZE(+1, +10)$			-0.007		0.005		-0.002			
			-0.67		0.58		-0.13			
BTM			-0.007		0.010		0.038			
			-0.23		0.94		0.83			
R^2	0.016		0.018		0.003		0.010		0.002	
Obs	726		486		1648		486		1648	

This table reports the results of predictive regressions after the announcement dates of dividend initiations and seasoned equity offerings (SEOs) over the sample period from 1983 to 2013. Panels A and C report regressions of the post-announcement return on the average daily conditional probabilities over the days after the dividend initiation date (DID) for NYSE/AMEX-listed target firms, and Panels B and D do the same after the SEO date (SEOD). The dependent variable, $R(+11, +60)$ or $R(+11, +30)$, is the average of daily returns from days +11 to +60 or +30 after the DID or the SEOD. Other

variables are defined as follows. $\pi_g(+1, +10)$: the average of daily π_g 's from day +1 to day +10 (relative to the DID or the SEOD), where π_g is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is good news; $\pi_b(+1, +10)$: the average of daily π_b 's from day +1 to day +10, where π_b is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is bad news; $R(+1, +10)$: the average of daily stock returns from day +1 to day +10; $RVOLA(+1, +10)$: the return volatility, which is the standard deviation of daily returns from day +1 to day +10; $PSPR(+1, +10)$: the average of PSPR from day +1 to day +10, where PSPR is the daily mean of intradaily proportional quoted spreads in % [i.e., ((dollar spread)/(quote mid-point))*100]; $OIMB(+1, +10)$: the average of OIMB from day +1 to day +10, where OIMB is the daily order imbalance in % [i.e., ((#BUY - #SELL)/(#BUY + #SELL))*100]; $TURN(+1, +10)$: the average of daily share turnover from day +1 to day +10; $SIZE(+1, +10)$: the natural logarithm of the average MV from day +1 to day +10, where MV is the daily market value (in \$million); and BTM : the book-to-market ratio in the previous quarter. The values in the first row for each explanatory variable are the coefficients from the regressions, and the values italicized in the second row of each variable are *t*-statistics. All coefficients are multiplied by 100. R^2 is the R-squared from the regressions. *Obs* is the number of target companies used in the regressions. Coefficients significantly different from zero at the significance levels of 1%, 5%, and 10% are indicated by ***, **, and *, respectively.

Table 8
Daily Posterior Probabilities of Informed Trading before Earnings Announcements

Mean Values of Conditional Probabilities in the 10 Portfolios Formed by Sorting on SUE Computed Based on EPS _{q-4} (1983:Q2-2013:Q4)												
Variable	SUE Portfolios										(High - Low)	
	Low					High						
	SUE1	SUE2	SUE3	SUE4	SUE5	SUE6	SUE7	SUE8	SUE9	SUE10	value	t-stat
SUE	-0.240	-0.013	-0.005	-0.001	0.001	0.002	0.004	0.006	0.012	0.140	0.380	6.03
$\pi_g(-5, -1)$	0.172	0.194	0.201	0.208	0.225	0.229	0.225	0.231	0.232	0.216	0.044	10.27
$\pi_b(-5, -1)$	0.130	0.134	0.140	0.146	0.142	0.145	0.139	0.133	0.111	0.063	-0.067	-18.71

This table reports the average of daily posterior probabilities of informed trading before the earnings announcement date (EAD) for portfolios formed by sorting on *SUE*. When the quarterly earnings were announced after the close (4:00 pm EST) on the IBES-reported announcement date, which is indicated by the time-stamps (available in IBES from 1999 on), the announcements are assumed to be made on the following trading day. *SUE* is the standardized unexpected earnings, which is defined by the quarterly earnings surprise computed based on the past (four quarters before) earnings per share (EPS_{q-4}) [following Abarbanell and Lehavy (2007), special items are excluded from the Compustat-reported EPS]. For each firm each quarter, the two daily posterior probabilities (π_g and π_b) are calculated for days around the quarterly EAD. To form decile portfolios, the component stocks are split into ten groups (with equal number of stocks) each quarter after being sorted in ascending order of *SUE*. In each portfolio in each quarter, the cross-sectional mean of the relevant probabilities obtained above is computed, and finally the time-series average of the quarterly cross-sectional means is reported. The variables are defined as follows. $\pi_g(-5, -1)$: the average of daily probabilities π_g from day -5 to day -1 relative to the EAD, where π_g is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that a good-news information event occurs on a given day; and $\pi_b(-5, -1)$: the average of daily probabilities π_b 's from day -5 to day -1, where π_b is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that a bad-news information event occurs on a given day. The last two columns contain the time-series average values for the differentials (*High - Low*) of *SUE* and the average probabilities between the highest *SUE* portfolio (SUE10) and the lowest *SUE* portfolio (SUE1), together with the *t*-statistics to test the hypothesis that the time-series average of the differences equals zero. The average number of component stocks used in each quarter is 1,202.4 (120.2 stocks in each portfolio). The sample period is from the second quarter of 1983 to the last quarter of 2013 (1983:Q2-2013:Q4) for NYSE/AMEX-listed stocks.

Table 9

The Proportions of Firms with Good- and Bad-News Days and the Differences of the Probabilities in the Most Positive (SUE10) and Negative (SUE1) Surprise Portfolios around the EAD

Proportions of the Daily Conditional Probabilities and their Differences around the EAD for SUE Portfolios												
Component	Category	Trading Days around the EAD										
		d - 5	d - 4	d - 3	d - 2	d - 1	d	d + 1	d + 2	d + 3	d + 4	d + 5
SUE10	$\pi_g \geq 0.9$	0.152	0.151	0.162	0.181	0.226	0.341	0.269	0.218	0.197	0.186	0.181
	$\pi_b \geq 0.9$	0.042	0.040	0.041	0.045	0.063	0.131	0.115	0.099	0.095	0.090	0.082
	$(\pi_g^{abn} - \pi_b^{abn})$	0.050	0.052	0.063	0.078	0.108	0.153	0.098	0.061	0.042	0.037	0.039
SUE1	$\pi_g \geq 0.9$	0.130	0.127	0.134	0.140	0.165	0.241	0.209	0.182	0.161	0.154	0.150
	$\pi_b \geq 0.9$	0.091	0.094	0.096	0.100	0.128	0.174	0.141	0.130	0.115	0.106	0.105
	$(\pi_g^{abn} - \pi_b^{abn})$	-0.002	-0.007	-0.002	-0.001	-0.005	0.022	0.024	0.012	0.006	0.007	0.006

This table reports the proportion of firms for which the daily conditional probabilities of informed trading are greater than or equal to 0.9 as well as the difference between the two abnormal probabilities ($\pi_g^{abn} - \pi_b^{abn}$) for the most positive (SUE10) and most negative (SUE1) surprise (measured by *SUE*) portfolios around the earnings announcement date (EAD). When the quarterly earnings were announced after the close (4:00 pm EST) on the IBES-reported announcement date, which is indicated by the time-stamps (available in IBES from 1999 on), the announcements are assumed to be made on the following trading days. *SUE* is the standardized unexpected earnings, which is defined by the quarterly earnings surprise computed based on the past (four quarters before) earnings per share (EPS_{q-4}) [following Abarbanell and Lehavy (2007), special items are excluded from the Compustat-reported EPS]. For each firm each quarter, the two daily conditional probabilities (π_g and π_b) are first assigned around the quarterly EAD. To form decile portfolios, the component stocks are split into ten groups (SUE1-SUE10) (with equal number of stocks) each quarter after being sorted in ascending order by *SUE*. The variables are defined as follows. π_g : the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that a good-news information event occurs on a given day; π_b : the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that a bad-news information event occurs on a given day; π_g^{abn} : the average of individual daily abnormal π_g 's; and π_b^{abn} : the average of individual daily abnormal π_b 's; and $(\pi_g^{abn} - \pi_b^{abn})$: the difference between the two abnormal probabilities. The abnormal probability for each component stock for each day around the event date (day *d*-5 to day *d*+5) is computed as the daily value of the probability (π_g or π_b) minus the average of the daily probabilities over the 30 pre-event days (from day *d*-40 to day *d*-11). The average number of component stocks used in each quarter is 1,202.4 (120.2 stocks in each portfolio). The sample period is from the second quarter of 1983 to the last quarter of 2013 for NYSE/AMEX-listed stocks.

Table 10
Pre-EAD Informed Trading and the Earnings Announcement Return (CAR)

Pre-Announcement Informed Trading around EAD and the Announcement Return (1983:Q2-2013:Q4)							
Dep. Var. = CAR(0, +1)							
Panel A: For SUE > 0			Panel B: For SUE < 0			Panel C: For All	
Expla. Variables	(i)	(ii)	Expla. Variables	(iii)	(iv)	Expla. Variables	(v)
Intercept	1.455 ***	1.444 ***	Intercept	-2.457 ***	-2.424 ***	Intercept	0.552 ***
	7.26	7.16		-9.82	-9.60		3.58
SUE	2.385 **	3.788 **	SUE	2.299 **	2.826 **	SUE	3.200 ***
	2.04	2.40		2.26	2.24		4.01
$\pi_g(-5, -1)$	-0.315 **	-0.318 **	$\pi_b(-5, -1)$	0.641 ***	0.527 ***	$\pi_i(-5, -1)$	-0.019
	-2.12	-1.96		3.79	2.80		-0.18
$SUE * \pi_g(-5, -1)$		-8.558 **	$SUE * \pi_b(-5, -1)$		-16.800 **	$SUE * \pi_i(-5, -1)$	-9.099 ***
		-2.29			-2.36		-3.85
R(-5, -1)	-33.184 ***	-32.934 ***	R(-5, -1)	-22.300 ***	-21.703 ***	R(-5, -1)	-25.371 ***
	-11.05	-11.02		-3.71	-3.50		-7.83
RVOLA(-5, -1)	3.500	3.419	RVOLA(-5, -1)	-0.187	-0.776	RVOLA(-5, -1)	3.158
	1.27	1.26		-0.06	-0.26		1.58
PSPR(-5, -1)	0.057	0.048	PSPR(-5, -1)	0.019	0.023	PSPR(-5, -1)	-0.007
	1.06	0.89		0.37	0.43		-0.20
OIMB(-5, -1)	-0.002	-0.003	OIMB(-5, -1)	-0.003 *	-0.002 *	OIMB(-5, -1)	-0.003 ***
	-1.31	-1.45		-1.80	-1.72		-2.84
TURN(-5, -1)	-20.410 ***	-19.745 ***	TURN(-5, -1)	17.099	14.757	TURN(-5, -1)	-8.794
	-3.03	-2.88		1.60	1.36		-1.44
SIZE(-5, -1)	-0.129 ***	-0.130 ***	SIZE(-5, -1)	0.230 ***	0.233 ***	SIZE(-5, -1)	0.004
	-5.78	-5.83		7.93	7.99		0.23
BTM	0.231 ***	0.236 ***	BTM	0.235 ***	0.205 **	BTM	0.255 ***
	2.91	2.92		2.63	2.25		4.17
						D_{sue}	-1.539 ***
							-20.36
Avg R ²	0.034	0.036	Avg R-sqr	0.044	0.046	Avg R-sqr	0.040
Avg Obs	651.6	649.4	Avg Obs	409.3	407.9	Avg Obs	1063.2

(Table 10: continued)

Pre-Announcement Informed Trading around EAD and the Announcement Return (1983:Q2-2013:Q4)								
Dep. Var. = $CAR(-1, +1)$								
Panel D: For $SUE > 0$			Panel E: For $SUE < 0$			Panel F: For All		
Expla. Variables	(vi)	(vii)	Expla. Variables	(viii)	(ix)	Expla. Variables	(x)	
Intercept	1.448 ***	1.425 ***	Intercept	-2.567 ***	-2.556 ***	Intercept	0.654 ***	
	6.24	6.11		-9.01	-8.97		3.41	
SUE	2.359 *	4.388 ***	SUE	2.194 **	2.714 **	SUE	3.902 ***	
	1.94	2.67		2.11	2.16		4.66	
$\pi_g(-5, -1)$	-0.591 ***	-0.530 ***	$\pi_b(-5, -1)$	0.550 ***	0.494 **	$\pi_l(-5, -1)$	-0.154	
	-3.70	-3.19		2.83	2.35		-1.34	
$SUE * \pi_g(-5, -1)$		-12.987 ***	$SUE * \pi_b(-5, -1)$		-10.704 *	$SUE * \pi_l(-5, -1)$	-11.017 ***	
		-3.05			-1.69		-4.51	
$R(-5, -1)$	79.065 ***	79.520 ***	$R(-5, -1)$	87.410 ***	87.725 ***	$R(-5, -1)$	85.894 ***	
	18.31	18.45		13.29	13.00		19.78	
$R(-5, -1)$	14.008 ***	14.264 ***	$R(-5, -1)$	-11.250 ***	-11.766 ***	$R(-5, -1)$	3.573	
	3.68	3.77		-2.93	-3.07		1.30	
$PSPR(-5, -1)$	0.026	0.016	$PSPR(-5, -1)$	0.085	0.089	$PSPR(-5, -1)$	0.003	
	0.45	0.27		1.36	1.41		0.08	
$OIMB(-5, -1)$	-0.005 ***	-0.005 ***	$OIMB(-5, -1)$	-0.005 ***	-0.005 ***	$OIMB(-5, -1)$	-0.006 ***	
	-2.61	-2.78		-3.25	-3.20		-4.76	
$TURN(-5, -1)$	-36.174 ***	-36.385 ***	$TURN(-5, -1)$	17.106	16.967	$TURN(-5, -1)$	-20.926 ***	
	-4.22	-4.16		1.27	1.26		-2.69	
$SIZE(-5, -1)$	-0.136 ***	-0.136 ***	$SIZE(-5, -1)$	0.253 ***	0.257 ***	$SIZE(-5, -1)$	0.003	
	-5.39	-5.48		7.41	7.48		0.12	
BTM	0.272 ***	0.278 ***	BTM	0.194 *	0.185 *	BTM	0.259 ***	
	3.11	3.15		1.94	1.83		3.88	
						D_sue	-1.735 ***	
							-24.51	
Avg R ²	0.070	0.071	Avg R ²	0.077	0.079	Avg R ²	0.077	
Avg Obs	651.6	649.4	Avg Obs	409.3	407.9	Avg Obs	1063.2	

This table reports the results of Fama and MacBeth(1973) cross-sectional regressions (for positive and negative SUE separately), run quarterly, using the variables computed around the earnings announcement date (EAD). Panels A and D report results for firms with positive SUE , Panels B and E for firms with negative SUE , and Panels C and F for all firms. The dependent variable is $CAR(0, +1)$ or $CAR(-1, +1)$, the cumulative abnormal return over two or three trading days (days 0 or -1 to +1) around the quarterly EAD. The explanatory variables are defined as follows. SUE : the standardized unexpected earnings, which is defined by the quarterly earnings surprise computed by assuming, for forecasting purposes, that the earnings per share (EPS) follows a seasonal random walk process [following Abarbanell and Lehavy (2007), special items are excluded from the Compustat-reported EPS]; $\pi_g(-5, -1)$: the average of daily probabilities π_g 's from day -5 to day -1 (relative to the EAD),

where π_g is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is good news; $SUE*\pi_g(-5, -1)$: the interaction term [SUE times $\pi_g(-5, -1)$]; $\pi_b(-5, -1)$: the average of daily probabilities π_b 's from day -5 to day -1, where π_b is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is bad news; $R(-5, -1)$: the average of daily stock returns from day -5 to day -1; $\pi_i(-5, -1)$: $\pi_g(-5, -1)$ if $SUE > 0$ and $\pi_b(-5, -1)$ if $SUE < 0$; $SUE*\pi_i(-5, -1)$: the interaction term [SUE times $\pi_i(-5, -1)$]; $RVOLA(-5, -1)$: the return volatility, which is the standard deviation of daily returns from day -5 to day -1; $PSPR(-5, -1)$: the average of PSPR from day -5 to day -1, where PSPR is the daily mean of intradaily proportional quoted spreads in % [i.e., $((\text{dollar spread})/(\text{quote mid-point}))*100$]; $OIMB(-5, -1)$: the average of OIMB from day -5 to day -1, where OIMB is the daily order imbalance in % [i.e., $((\#BUY - \#SELL)/(\#BUY + \#SELL))*100$]; $TURN(-5, -1)$: the average of daily share turnover from day -5 to day -1; $SIZE(-5, -1)$: the natural logarithm of the average MV from day -5 to day -1, where MV is the daily market value (in \$million); BTM : the book-to-market ratio in the previous quarter; and D_{SUE} : 1 if $SUE < 0$, and 0 otherwise. The values in the first row for each explanatory variable are time-series averages of coefficients obtained from the quarterly cross-sectional regressions, and the values italicized in the second row of each variable are t -statistics computed based on Fama-MacBeth (1973). All coefficients are multiplied by 100. *Avg R²* is the average of adjusted R-squareds from quarterly regressions. *Avg Obs* is the average number of companies used each quarter in the cross-sectional regressions. Coefficients significantly different from zero at the significance levels of 1%, 5%, and 10% are indicated by ***, **, and *, respectively. The sample period is from the second quarter of 1983 to the last quarter of 2013 (1983:Q2-2013:Q4) for NYSE/AMEX stocks.

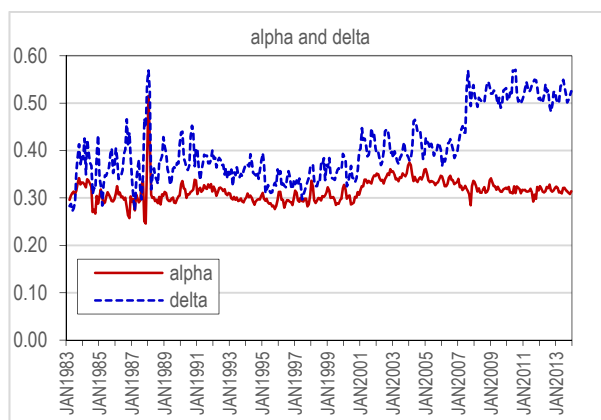
Table 11
Post-EAD Informed Trading and Subsequent Stock Returns

Predictive Regressions after Earnings Announcements: Dep. Var. = $CAR(+3, +15)$		
Explana. Var.	(i)	(ii)
Intercept	0.110	0.164
	<i>0.40</i>	<i>0.35</i>
$\pi_g(+1, +2)$	0.579 ***	0.598 ***
	<i>3.38</i>	<i>4.06</i>
$\pi_b(+1, +2)$	-0.592 ***	-0.344 *
	<i>-3.27</i>	<i>-1.93</i>
$CAR(+1, +2)$		-4.568 ***
		<i>-2.91</i>
$RVOLA(+1, +2)$		2.940
		<i>0.87</i>
$PSPR(+1, +2)$		0.407 ***
		<i>4.97</i>
$OIMB(+1, +2)$		0.003
		<i>1.13</i>
$TURN(+1, +2)$		14.568
		<i>1.23</i>
$SIZE(+1, +2)$		-0.071
		<i>-1.16</i>
BTM		-0.575 ***
		<i>-4.49</i>
Avg R^2	0.003	0.024
Avg Obs	1095.0	1028.6

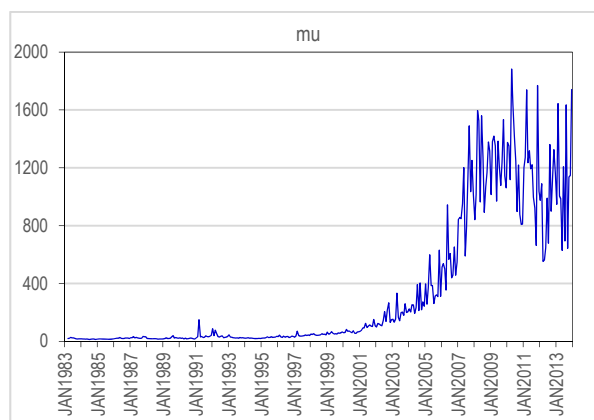
This table reports the results of quarterly Fama-MacBeth (1973) cross-sectional regressions of the cumulative abnormal return over days +3 to +15 after the earnings announcement date (EAD) on the probabilities of informed trading after the EAD (from day +1 to day +2) for NYSE/AMEX-listed firms. The sample period is from 1983 to 2013. The dependent variable is $CAR(+3, +15)$, the cumulative abnormal return over trading days from day +3 to day +15 after the EAD. Other variables are defined as follows. $\pi_g(+1, +2)$: the average of daily π_g 's from day +1 to day +2, where π_g is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is good news; $\pi_b(+1, +2)$: the average of daily π_b 's from day +1 to day +2, where π_b is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is bad news; $CAR(+1, +2)$: the cumulative abnormal return over two trading days (days +1 and +2) after the EAD; $RVOLA(+1, +2)$: the return volatility, which is the standard deviation of daily returns from day +1 to day +2; $PSPR(+1, +2)$: the average of PSPR from day +1 to day +2, where PSPR is the daily mean of intradaily proportional quoted spreads in % [i.e., $((\text{dollar spread})/(\text{quote mid-point})) \times 100$]; $OIMB(+1, +2)$: the average of OIMB from day +1 to day +2, where OIMB is the daily order imbalance in % [i.e., $((\#BUY - \#SELL)/(\#BUY + \#SELL)) \times 100$]; $TURN(+1, +2)$: the average of daily share turnover from day +1 to day +2; $SIZE(+1, +2)$: the natural logarithm of the average MV from day +1 to day +2, where MV is the daily market value (in \$million); and BTM : the book-to-market ratio in the previous quarter. The values in the first row for each explanatory variable are the average coefficients from the quarterly regressions, and the values italicized in the second row of each variable are Fama-MacBeth (1973) t -statistics. All coefficients are multiplied by 100. Avg R^2 is the average of adjusted R-squareds from the quarterly regressions. Avg Obs is the average number of companies used each quarter in the regressions. Coefficients significantly different from zero at the significance levels of 1%, 5%, and 10% are indicated by ***, **, and *, respectively.

Figure 1. Time-Series Plots for the Monthly Cross-Sectional Averages of the *PIN*-related Parameters

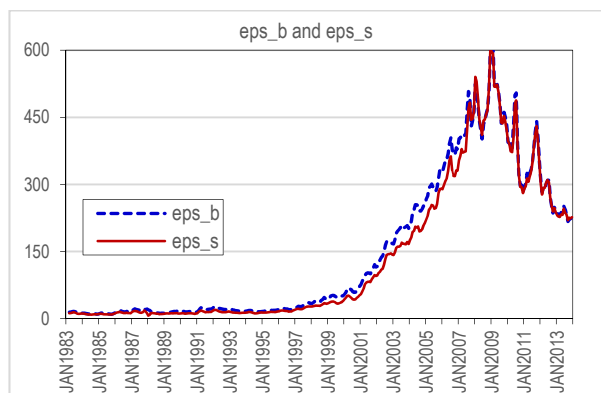
This figure plots the time series of the equal-weighted cross-sectional mean of the *PIN*-related parameters estimated on a monthly basis for NYSE/AMEX-listed stocks over the 31 years (1983:03-2013:12). The sample includes common stocks only (SHRCD = 10 or 11 in CRSP). Figure 1(A) shows the time-series plot for α and δ , Figure 1(B) does the same for μ , and Figure 1(C) does the same for ϵ_B and ϵ_S . The five *PIN*-related parameters are defined as follows: α (*alpha* in the legend) is the probability with which a private information event occurs on a given day; and δ (*delta* in the legend) is the probability with which a private information event, if it occurs on a given day, contains bad news; μ (*mu* in the legend) is the rate at which orders from informed traders arrive if the information event does occur; ϵ_B (*eps_b* in the legend) is the rate at which orders from uninformed buyers arrive; and ϵ_S (*eps_s* in the legend) is the rate at which orders from uninformed sellers arrive. For estimating the monthly *PIN*-related parameters, each trade in the ISSM/TAQ databases is classified as buyer-initiated or seller-initiated via the Lee-Ready (1991) algorithm up to 2006 and the Holden-Jacobsen (2014) algorithm for the 2007-2013 period. The average number of component stocks used in each month is 1,876.8.



< Figure 1(A): Average of α and δ >



< Figure 1(B): Average of μ >



< Figure 1(C): Average of ϵ_B and ϵ_S >

Figure 2

Time-Series Plots for the Monthly Cross-Sectional Averages of the Daily Posterior Probabilities

This figure plots the time series of the equal-weighted monthly averages of the two daily posterior probabilities for NYSE/AMEX stocks over the 31 years (1983:04-2013:12). Each day, π_g and π_b are estimated for each stock using the daily aggregated numbers of buyer-initiated trades and seller-initiated trades. For each firm, π_g and π_b are first averaged across trading days within each month, and then the cross-sectional mean of the monthly averages is calculated each month. The average number of component stocks used in each month is 1,876.8.

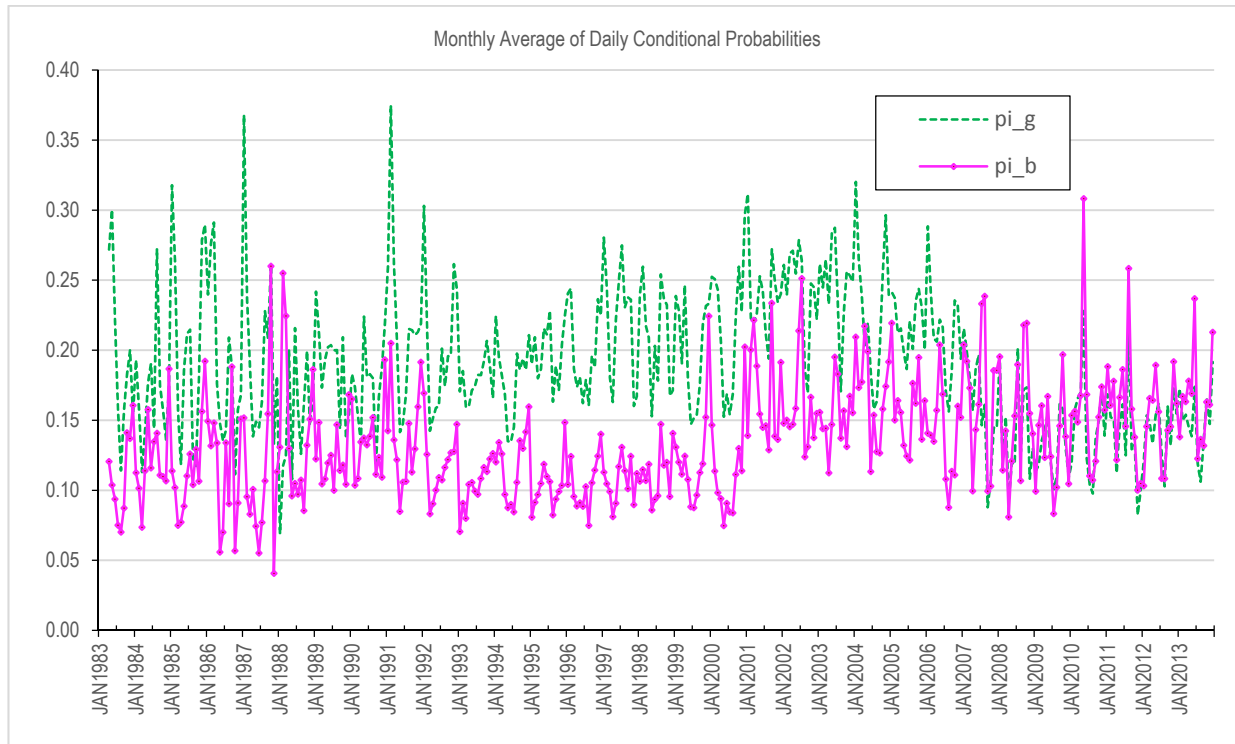


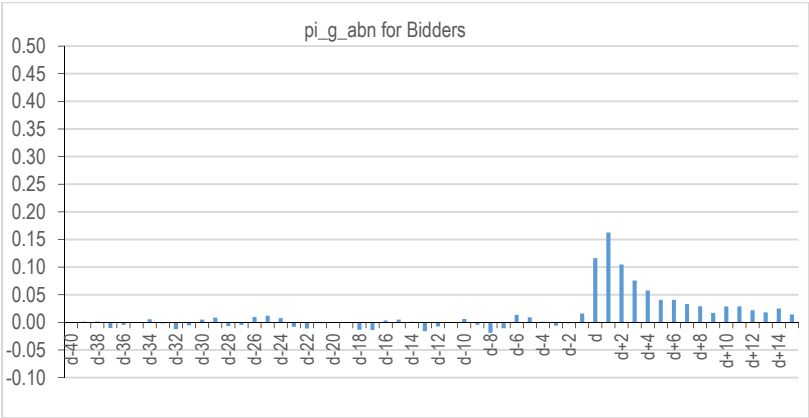
Figure 3

Daily Abnormal Probabilities around the M&A Announcement Date for Bidder and Target Firms

The figure plots the averages of daily abnormal probabilities (π_g^{abn} and π_b^{abn}) around the mergers and acquisitions (M&A) announcement date (M&AD: day d) for bidding (or acquiring) firms in Panel A and for target firms in Panel B. The abnormal probability for each component stock for each day around the event date (day $d-40$ to day $d+15$) is computed as the daily value of the probability (π_g or π_b) minus the average of the daily probabilities over the 20 days from day $d-60$ to day $d-41$. In Panel A, the average number of component stocks is 7,172 NYSE/AMEX-listed bidding firms for the 1983-2013 period. In Panel B, the average number of component stocks is 2,623 NYSE/AMEX-listed target firms for the 1983-2013 period. The M&A-related variables are extracted from the SDC Platinum database for both completed and withdrawn M&A deals (tender offers) whose transaction values are greater than \$50 million, and then matched with CRSP daily returns and other identification variables as well as the daily conditional probabilities estimated as in Table 1. The definitions of the variables are as follows. π_g^{abn} : the average of individual daily abnormal π_g 's, where π_g is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is good news; and π_b^{abn} : the average of individual daily abnormal π_b 's, where π_b is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is bad news.

(Figure 3: continued)

Panel A: Bidding Firms



Panel B: Target Firms

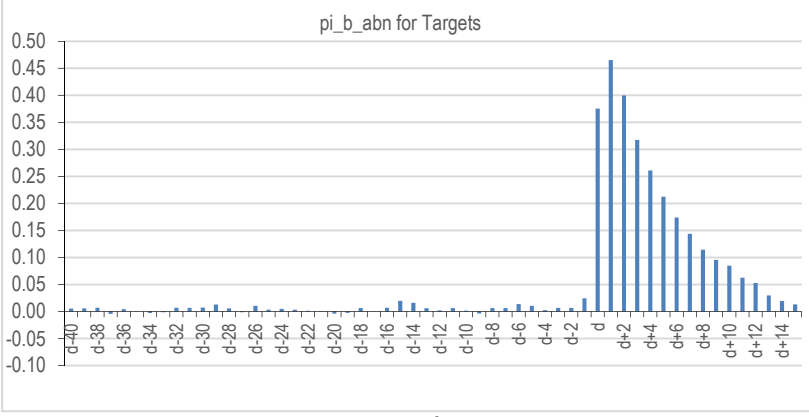
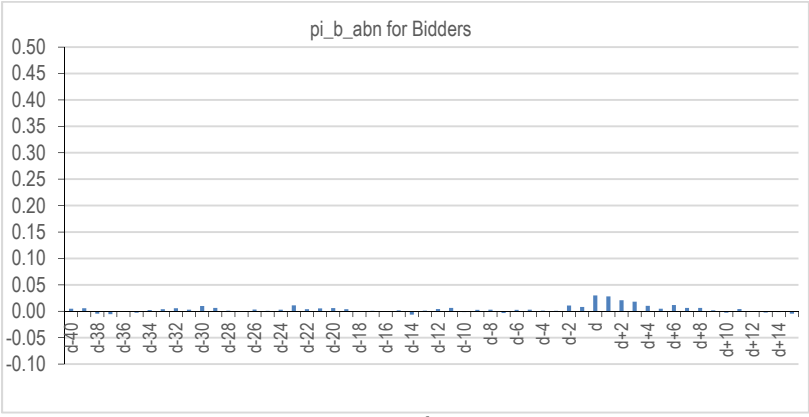
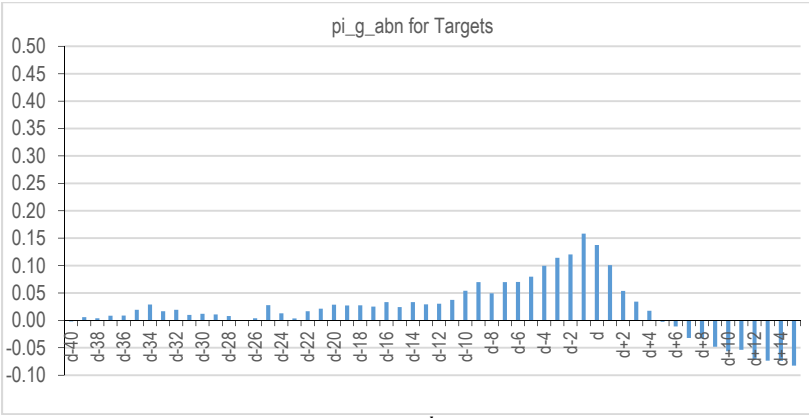
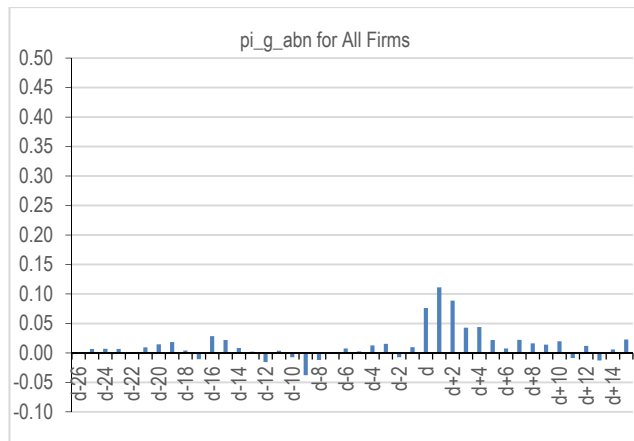


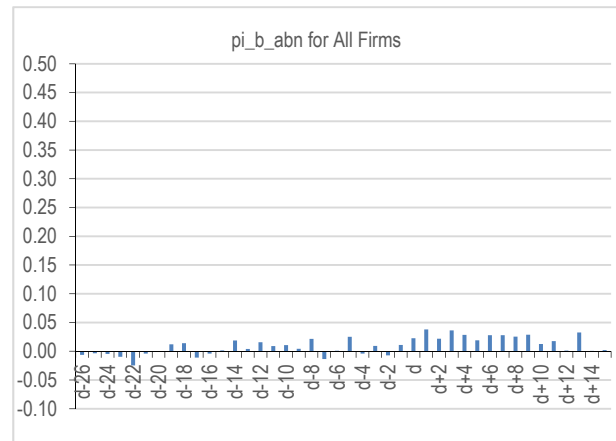
Figure 4

Daily Abnormal Posterior Probabilities of Informed Trading around the Dividend Initiation Date

The figure plots the averages of daily abnormal probabilities (π_g^{abn} and π_b^{abn}) around the dividend initiation date (DID: day d) for all firms available in our sample. DID is defined as the date on which cash dividend is declared for the first time in a firm's history. The dividend initiation sample includes firms that pay ordinary quarterly dividend only (i.e., distribution code = 1234 in Compustat). With the above constraints, 726 firms are available from Compustat. The abnormal probability for each component stock for each day around the event date is computed as the daily value of the probability (π_g or π_b) minus the average of the corresponding daily probabilities over the 20 days from day $d-60$ to day $d-41$. The definitions of the variables are as follows. π_g^{abn} : the average of individual daily abnormal π_g 's, where π_g is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is good news; π_b^{abn} : the average of individual daily abnormal π_b 's, where π_b is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is bad news. The sample period is from 1983 to 2013 for NYSE/AMEX-listed stocks.



< Figure 4(A): Average π_g^{abn} around DID for All Firms >



< Figure 4(B): Average π_b^{abn} around DID for All Firms >

Figure 5**Daily Abnormal Posterior Probabilities of Informed Trading around the Seasoned Equity Offerings Date**

The figure plots the averages of daily abnormal probabilities (π_g^{abn} and π_b^{abn}) around the seasoned equity offerings date (SEOD: day d) for 1,714 SEO cases. SEOD is defined as the filing date of SEOs in the Global New Issues database of SDC Platinum. To be included in our SEO sample, a firm should file at least \$25 million and offer at least some primary shares. The abnormal probability for each component stock for each day around the event date is computed as the daily value of the probability (π_g or π_b) minus the average of the corresponding daily probabilities over the 20 days from day $d-60$ to day $d-41$. The definitions of the variables are as follows. π_g^{abn} : the average of individual daily abnormal π_g 's, where π_g is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is good news; π_b^{abn} : the average of individual daily abnormal π_b 's, where π_b is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is bad news. The sample period is from 1983 to 2013 for NYSE/AMEX-listed stocks.

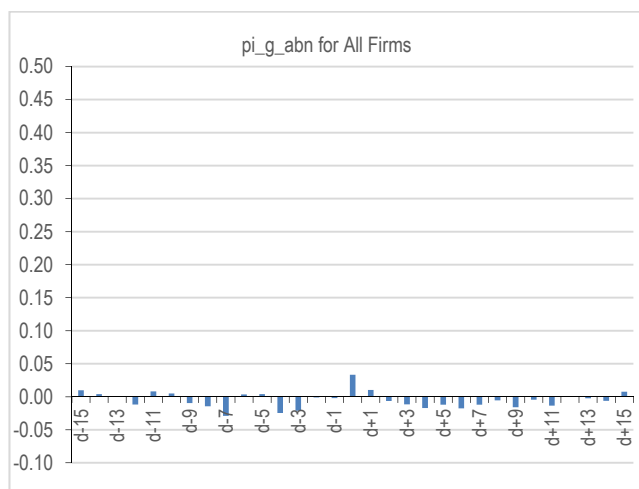
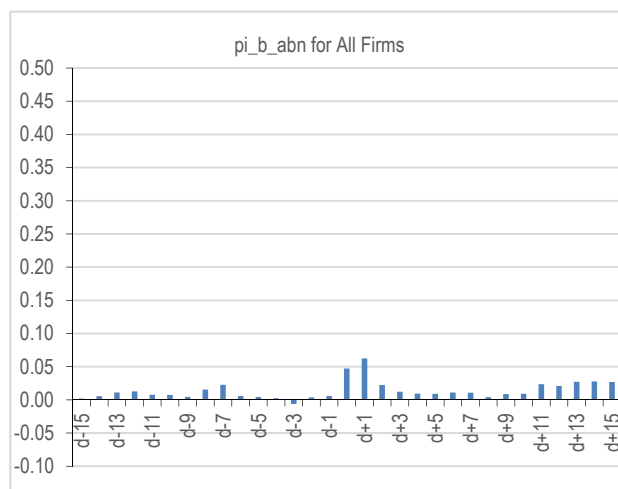
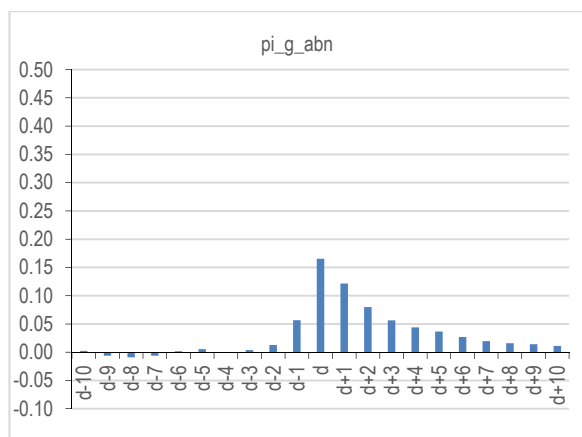
< Figure 5(A): Average π_g^{abn} around SEOD for All Firms >< Figure 5(B): Average π_b^{abn} around SEOD for All Firms >

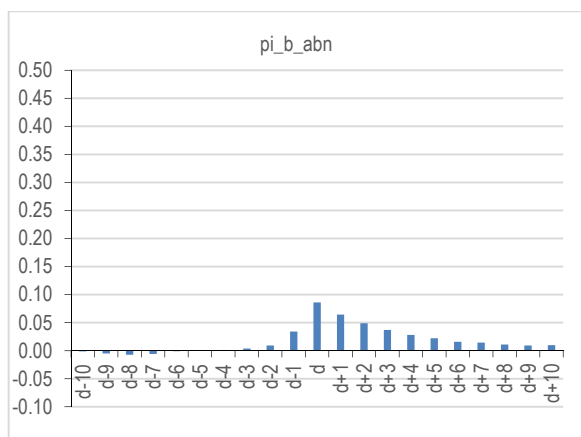
Figure 6

Daily Abnormal Posterior Probabilities of Informed Trading around the Quarterly Earnings Announcement Date

The figure plots the averages of daily abnormal probabilities (π_g^{abn} and π_b^{abn}) around the quarterly earnings announcement date (EAD: day d). The abnormal probability for each component stock for each day around the event date (day $d-10$ to day $d+10$) is computed as the daily value of the probability (π_g or π_b) minus the average of the corresponding daily probabilities over the 30 days from day $d-40$ to day $d-11$. The definitions of the variables are as follows. π_g^{abn} : the average of individual daily abnormal π_g 's, where π_g is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is good news; π_b^{abn} : the average of individual daily abnormal π_b 's, where π_b is the daily posterior probability (conditional on observing the number of daily buyer- and seller-initiated trades) that the private information is bad news. The average number of component stocks used in each quarter is 1,412.0. The sample period is from the second quarter of 1983 to the last quarter of 2013 (1983:Q2-2013:Q4) for NYSE/AMEX stocks.



< Figure 6(A): Average π_g^{abn} around EAD >



< Figure 6(B): Average π_b^{abn} around EAD >